

REPORT

Life without light: ocean trenches, chemosynthesis and the role of CO_2

A fact-checked synthesis of the scientific literature on the deepest ecosystems in the ocean

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Version 6.3 · October 2026 · English translation of the Spanish original

23 original figures · 3 photographs · 6 scientific plates

Cover image: chimneys at the Champagne site, NW Eifuku volcano (Mariana Arc). Pacific Ring of Fire 2004 Expedition, NOAA Office of Ocean Exploration. Public domain.

Executive summary

Purpose. This report brings together and checks what is known, as of October 2026, about life in ocean trenches, the chemosynthesis that sustains it and the role CO_2 plays in that system.

Method. A synthesis of published scientific literature. Every claim carries a confidence level and a note on whether it was checked against a source. It is not original research.

Main findings:

1. **There are animal ecosystems based on chemosynthesis down to 9,533 m**, spread along 2,500 km of the Kuril–Kamchatka and Aleutian trenches (*Nature*, 2025). They exceed the previous record by more than 2,000 m.
2. **CO_2 runs through the whole system**: buried organic matter releases CO_2, microbes turn it into methane, the methane rises along faults, is oxidized, and returns inorganic carbon that symbiotic bacteria fix again.
3. **Trenches concentrate life rather than exclude it**: their sediments are 2 to 8 times more active than those of the neighboring plains and bury carbon about 70 times faster than the deep-sea average.
4. **Microbial biodiversity is mostly unknown**: of 7,564 species found in the Mariana Trench, 89.4% were new.
5. **There is a biochemical depth limit for fish** at around 8,200–8,400 m; the deepest fish ever filmed was at 8,336 m.
6. **Human pollutants reach the bottom**: PCBs, mercury, carbon from nuclear tests and microplastics.
7. **"Without light" does not mean "without the Sun"**: almost all of this life depends on oxygen and organic matter produced by photosynthesis. Only systems fed by geological hydrogen, such as Lost City, are independent of it.
8. **"Dark oxygen" has not been confirmed**: nobody has observed it independently, and the original paper has carried an editor's note since April 2026.

What remains open. Section 17.2 reviews nine questions and what science can say about each today. The most important: whether these communities exist in other trenches, how much methane lies beneath them, and what share of hadal fauna really lives on chemosynthesis.

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How to read this report

A popular-science report written with care for accuracy. Every piece of data carries a confidence level between 0.0 and 1.0 and an origin tag:

- **[V]** = checked against a source while the report was being written (see References).
- **[C]** = general knowledge of the scientific literature, not re-checked item by item.

Each technical term is explained the first time it appears and is listed, with its etymology, in the final Glossary. This English version is a translation of the Spanish original.

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1. Introduction

Ocean trenches are the deepest and least explored places on the planet: narrow valleys between 6 and 11 km deep, in darkness, a few degrees above freezing and under pressures of up to 1,100 atmospheres. For a long time they were thought to be almost lifeless.

That picture has changed. Two results published in 2025 force a rethink: whole communities of tubeworms and clams below 9,500 m in the northwest Pacific, and thousands of unknown microbial species in the Mariana Trench. Both point the same way: in trenches, life does not merely use what sinks from above; it also makes food with chemical energy, and CO_2 is involved at every step of that process.

Scope. The report explains that system for a reader with no specialist training: where it happens, what chemistry drives it, what has been found and what is still unknown. It brings together published scientific literature and separates, throughout, what has been checked from what is doubtful.

Organization:

Sections	Content
2 to 4	The setting: the layers of the ocean, the trenches and what the seafloor feeds on
5 to 8	The chemistry: what chemosynthesis is, its reactions, the kinds of oasis and the role of CO_2
9 to 12	What has been found: the 2025 discovery, the fauna, its adaptations and the human footprint
13 to 15	Reliability: what has not been demonstrated, the implications and the verification table
16 to 19	Glossary, limitations, open questions, conclusions, licenses and references

A caution from the outset. "Without light" does not mean "without the Sun". Almost all the life described here depends, indirectly, on products of photosynthesis. This is the nuance most often left out when the subject is popularized, and section 13 develops it.

2. The ocean in layers

The ocean is divided into zones by depth. The names come from Greek and describe exactly what they are.

Zone (water column)	Depth	Light	Origin of the name
Epipelagic (photic)	0–200 m	Enough for photosynthesis	*epi* "upon" + *pélagos* "open sea"
Mesopelagic (twilight)	200–1,000 m	Dim; photosynthesis not possible	*mesos* "middle"
Bathypelagic (midnight)	1,000–4,000 m	None	*bathýs* "deep"
Abyssopelagic (abyssal)	4,000–6,000 m	None	*ábyssos* "bottomless"
Hadopelagic (hadal)	6,000–~11,000 m	None	*Hades*, Greek god of the underworld

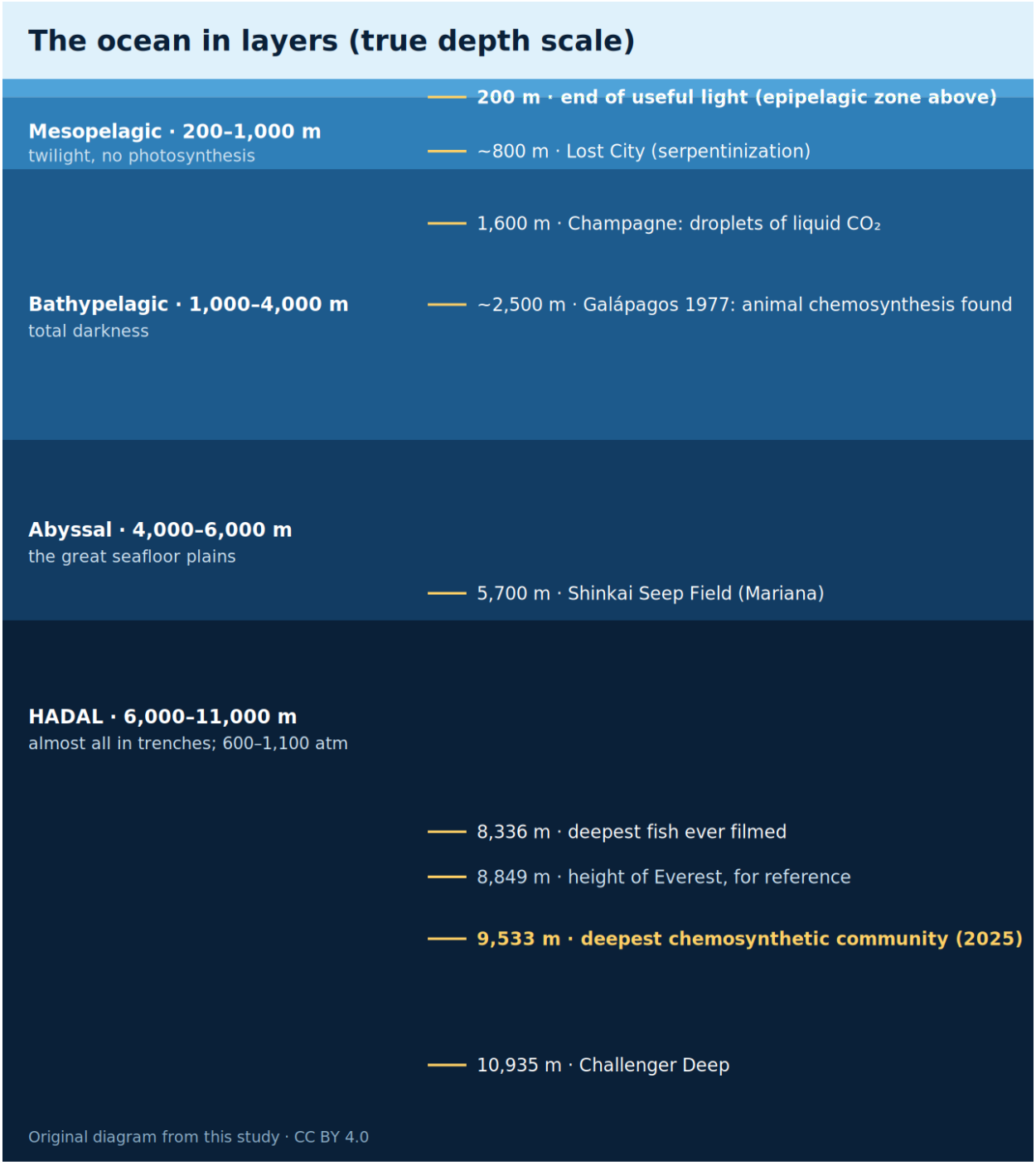


Figure 1. Ocean zones at true depth scale, with the places and records mentioned in this report.

Confidence in the boundaries: 0.9 **[C]**. They are conventions; different authors shift the limits by a few hundred meters.

Two words used throughout:

- **Pelagic**: living or happening in the water column, away from the bottom.
- **Benthic** (*bénthos*, "bottom"): living on or in the seafloor. The communities described in this report are benthic.

The lower limit of "hadal" life was set at 6,000 m in 1960, based on the fauna known at the time. **[V]** (0.85)

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3. Trenches: geology and physical conditions

3.1 How a trench forms

An **ocean trench** is a long, narrow depression with a V-shaped cross-section. It forms by **subduction** (Latin *sub* "under" + *ducere* "to lead"): a dense tectonic plate sinks beneath a less dense one. **[V]** (0.95)

Geological features that matter for biology:

- **Accretionary prism**: a wedge of sediment that the sinking plate scrapes off and piles up against the upper plate. *Accretion* means growth by addition. As it is compressed, that sediment expels fluids.
- **Faults**: fractures in the crust. They act as pipes along which those methane- and sulfide-laden fluids rise.
- **Forearc**: the region between the trench and the chain of volcanoes that subduction produces.
- **Funnel effect**: the V shape and earthquakes channel sediment and organic matter toward the axis. The floor of a trench receives more food than the surrounding abyssal plain.

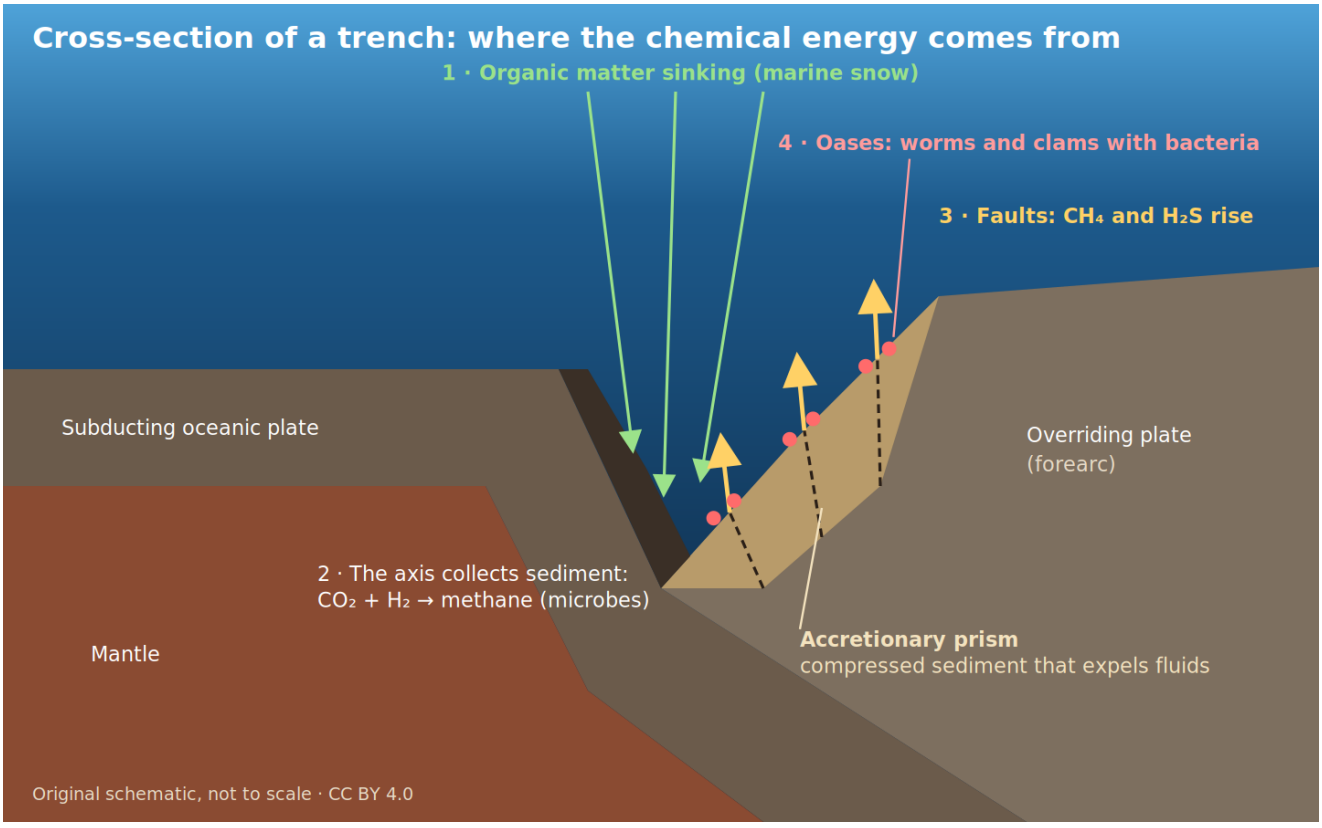


Figure 2. Schematic cross-section of a trench (not to scale): organic matter falls to the axis, is turned into methane beneath the sediment, and fluids rise along faults to the oases.

3.2 How many there are, and where

Item	Value	Confidence
Hadal habitats on the planet	46: 33 trenches (27 subduction trenches and 6 trench faults) and 13 troughs	0.85 **[V]**
Distribution of the 33 trenches	26 in the Pacific, 3 in the Atlantic, 2 in the Indian Ocean, 2 in the Southern Ocean	0.85 **[V]**

Item	Value	Confidence
Mean depth of the trenches	8,216 m	0.85 **[V]**
Trenches deeper than 10,000 m	Mariana, Tonga, Kuril–Kamchatka, Philippine and Kermadec	0.9 **[V]**
Share of the ocean's depth range	45%	0.9 **[V]**
Share of the seafloor area	**Disputed figure**: sources give anything from under 0.2% to 1–2%	0.5 **[V]**

> **Finding of the fact-check.** The figure for hadal area is not consistent between sources: a text by Alan Jamieson gives less than 0.2%, the MEER project portal says less than 1%, and a 2025 university press release says 1–2%. The difference depends on what is counted (trenches only, or also troughs and fracture zones) and on the limit used (6,000 or 6,500 m). The figure of 45% of the vertical range is consistent.

3.3 Pressure, temperature and darkness

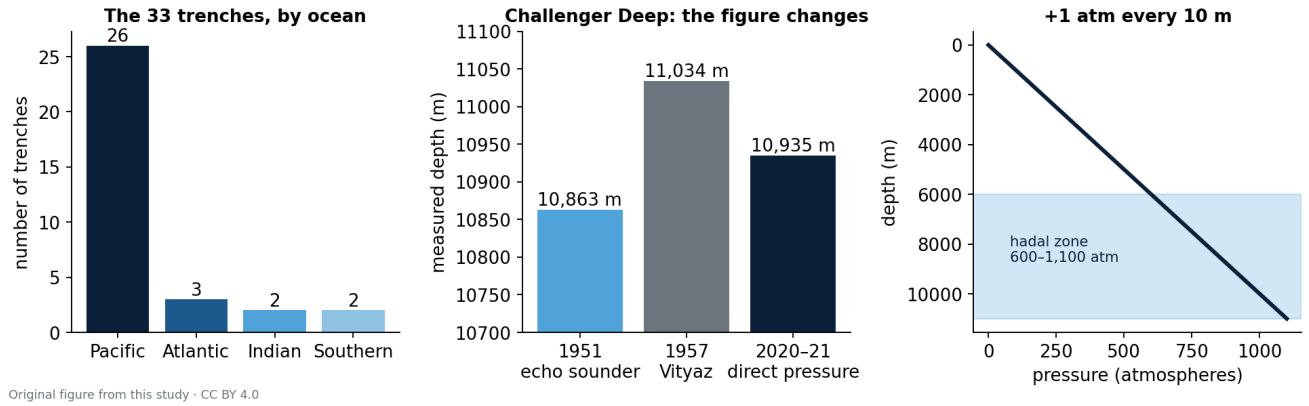
- **Hydrostatic pressure**: the weight of the water column. It rises by 1 atmosphere for every 10 m. **[V]** (0.95). At 11,000 m it is about 1,100 atm, that is, about 110 megapascals (MPa) or 1.1 metric tons on every square centimeter. In the hadal zone it ranges from about 600 to 1,100 atm. **[V]**
- **Temperature**: 1–4 °C at the bottom of a typical trench. **[V]** (0.9)
- **Light**: none from the Sun. The only light is **bioluminescence** (light produced by living organisms through a chemical reaction).

3.4 The deepest point

Challenger Deep, in the Mariana Trench, is the deepest known point. Its measurement history shows why there is no "exact" figure:

Year	Method	Depth
1875	Weighted sounding line, HMS *Challenger*	First measurement (gives the site its name)
1951	Echo sounder, HMS *Challenger II*	10,863 m
1957	Soviet vessel *Vityaz*	11,034 m (now considered an overestimate)
2020–2021	Direct pressure measurements	**10,935 m ± 6 m** (95% confidence interval)

Confidence: 0.85 **[V]**. Other papers cite 10,994 m. **Bathymetry** (the measurement of depths; *bathýs* + *métron* "measure") by **echo sounder** depends on the speed of sound in water, which changes with temperature, salinity and pressure.



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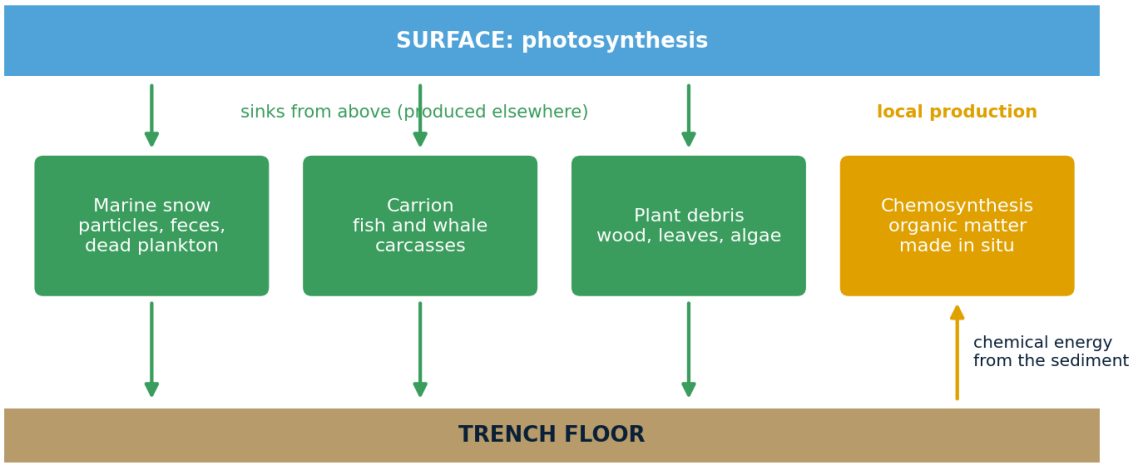
Figure 3. Left: the 33 trenches by ocean. Center: three measurements of Challenger Deep. Right: pressure rises by one atmosphere every 10 m.

4. Living without light: what the seafloor feeds on

With no local photosynthesis there are four food sources. The first three are remains of surface life; only the fourth is local production.

Source	What it is	Importance in trenches
Marine snow	A continuous rain of organic particles: dead plankton, fecal pellets, mucus	Classical base of the deep food chain
Carrion	Carcasses of fish, whales, squid	Large but unpredictable pulses
Plant debris	Wood, leaves, algae washed in from land	More important than once thought (see 10.2)
Chemosynthesis	Organic matter made in situ with chemical energy	Underestimated until 2025

Animals that eat carrion are called **scavengers**. A nutrient-poor environment is called **oligotrophic** (*olígos* "little" + *trophé* "food"); the hadal zone generally is, except along trench axes and at chemical oases.



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Figure 4. The four food sources of the hadal seafloor. Three arrive from the surface; only chemosynthesis is produced on the bottom itself.

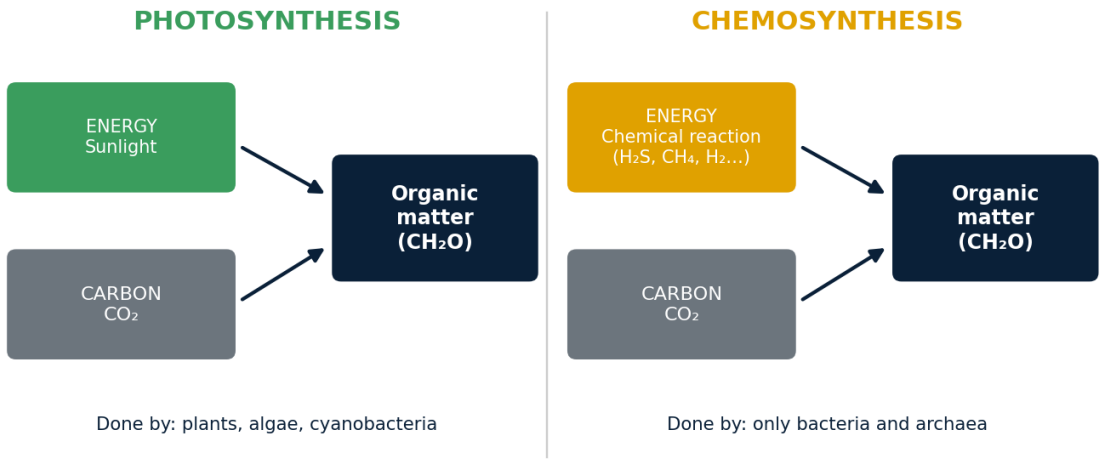
5. Chemosynthesis: history and fundamentals

5.1 Definition

Chemosynthesis (*chemeía* "chemistry" + *sýnthesis* "putting together"): the making of organic matter from inorganic carbon (CO_2) using the energy of chemical reactions instead of light.

Direct comparison:

	Photosynthesis	Chemosynthesis
Energy source	Light	Chemical reactions (oxidation of H_2S, CH_4, H_2, NH_3, Fe^{2+})
Carbon source	CO_2	CO_2 (or CH_4 in methanotrophs)
Who does it	Plants, algae, cyanobacteria	Only bacteria and archaea
Where	Where there is light	Where a reduced compound meets an oxidant



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Figure 5. The two ways of making organic matter from CO_2. Only the energy source changes.

5.2 Metabolic vocabulary

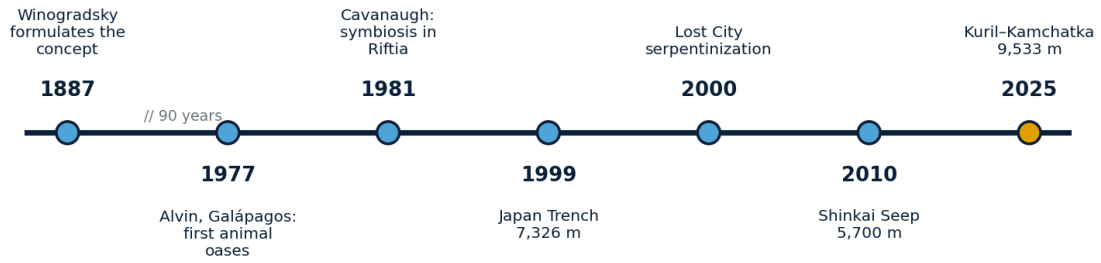
Microbiologists classify living things with three questions. Each answer is a prefix:

Question	Options	Etymology
Where does it get its **energy**?	**Photo**- (light) / **Chemo**- (chemistry)	
Where does it get its **electrons**?	**Litho**- (inorganic compounds) / **Organo**- (organic compounds)	*líthos* "stone"
Where does it get its **carbon**?	**Auto**- (CO_2) / **Hetero**- (ready-made organic matter)	*autós* "self", *héteros* "other"; *trophé* "food"

A bacterium that oxidizes sulfide and fixes CO_2 is, in full, a **chemolithoautotroph**: "one that feeds itself using the chemistry of stones". We are chemoorganoheterotrophs.

5.3 Timeline

Year	Event	Confidence
~1887	Sergei Winogradsky describes bacteria that live by oxidizing inorganic compounds and formulates the concept	0.9 **[C]**
February 1977	The submersible *Alvin* dives on the Galápagos Rift (~2,500 m) looking for hydrothermal vents. The expedition was made up of geologists, with no biologists on board. They find foot-and-a-half-long clams and giant tubeworms	0.9 **[V]**
1981	The worm *Riftia pachyptila* is described. Colleen Cavanaugh, then a graduate student, shows that it lives on chemosynthetic symbiotic bacteria housed inside its body	0.9 **[V]**
1984	The first cold seeps are discovered (Gulf of Mexico)	0.8 **[C]**
1999	Hadal chemosynthetic community in the Japan Trench at 7,326 m, found by the Japanese vehicle *Kaiko*	0.9 **[V]**
2000	Discovery of the Lost City hydrothermal field (Atlantis Massif, Atlantic), driven by serpentinization	0.95 **[V]**
2010	*Shinkai Seep Field*: a serpentinite-hosted ecosystem at ~5,700 m, 80 km from Challenger Deep	0.9 **[V]**
2025	Chemosynthetic communities down to 9,533 m (Kuril–Kamchatka and Aleutian trenches)	0.95 **[V]**



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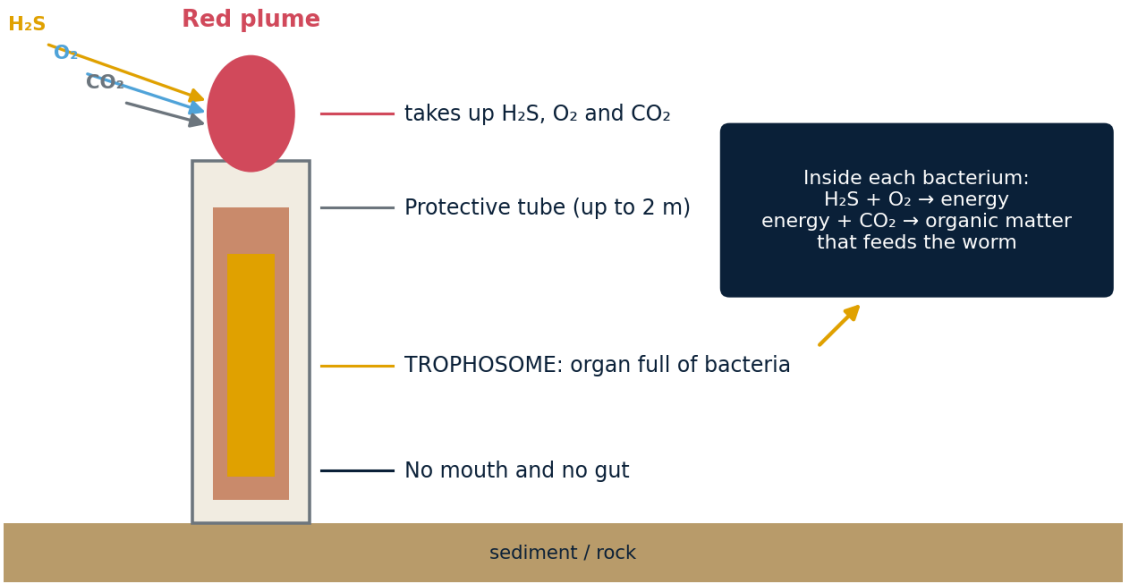
Figure 6. Milestones in the discovery of chemosynthetic life.

5.4 Symbiosis: the animals' trick

No animal performs chemosynthesis. What animals do is host bacteria that can. That is a **symbiosis** (*sýn* "with" + *bíos* "life"): two species living closely together. When both benefit it is called **mutualism**.

The founding example is *Riftia pachyptila* **[V]** (0.9):

- It has no mouth and no digestive tract.
- Its tubes reach 2 m in length.
- Inside it has an organ called the **trophosome** (*trophé* "food" + *sôma* "body") packed with bacteria. Because they live inside the animal's cells they are called **endosymbionts** (*éndon* "within").
- The worm takes up H_2S, O_2 and CO_2 with its red plume and carries them in its blood to the bacteria; the bacteria make organic matter and feed the worm.



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Figure 7. Schematic of a tubeworm such as *Riftia pachyptila*: the animal carries the gases and its internal bacteria do the chemistry.



Photo 1. *Riftia pachyptila* tubeworms on the Galápagos Rift: white tubes and red plumes. Image: NOAA Okeanos Explorer Program, Galápagos Rift Expedition 2011. Public domain.

In bivalves (clams and mussels) the bacteria live in the **gills**, inside specialized cells called **bacteriocytes**. The hadal clam *Maorithyas hadalis*, from the Japan Trench, hosts two different kinds of bacteria in different regions of the gill. **[V]** (0.85)

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6. The reactions, one by one

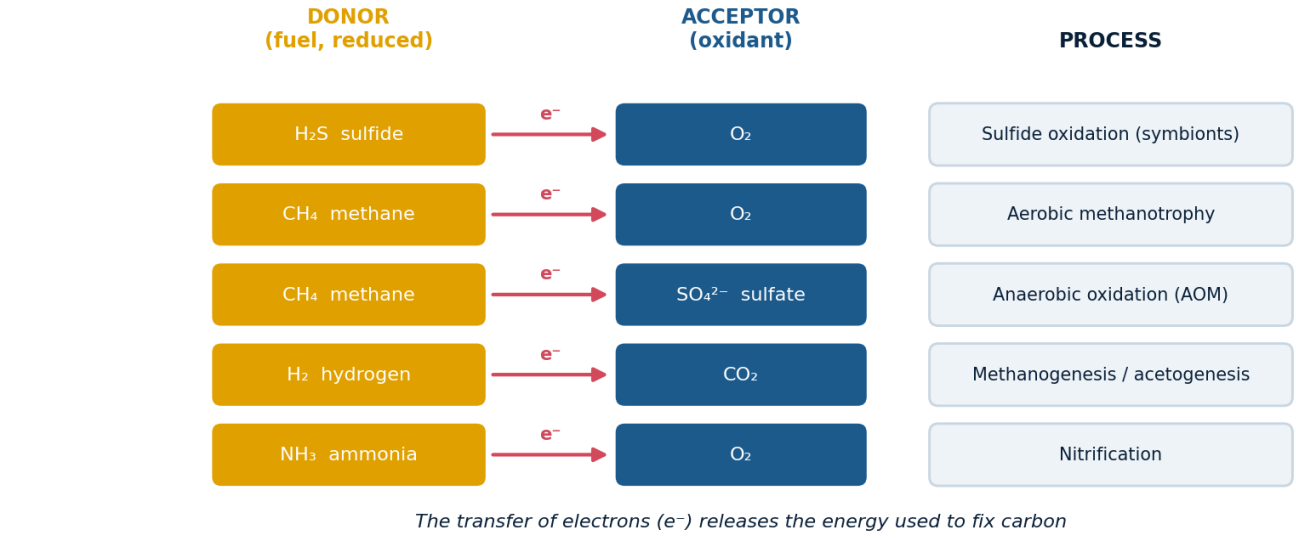
6.1 The central idea: redox reactions

All chemosynthesis rests on a **redox** reaction (reduction + oxidation): one substance gives up electrons and another accepts them. That transfer of electrons releases energy, as in a battery.

- **Electron donor** (the "fuel"): a **reduced**, electron-rich compound. Examples: H_2S, CH_4, H_2, NH_3.

- **Electron acceptor** (the "oxidant"): an **oxidized** compound that receives them. Examples: O_2, sulfate (SO_4^{2-}), nitrate (NO_3^-), CO_2.

Chemosynthetic life appears where the two meet: reduced fluid rising from the sediment and oxidized seawater coming down. That is why these ecosystems are narrow bands, and why they are called oases.



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Figure 8. The donor–acceptor pairs that sustain life without light.

6.2 Table of reactions

The equations are simplified and balanced. `CH₂O` stands for organic matter in general.

#	Process	Reaction	Who does it	Where
1	Sulfide oxidation with CO_2 fixation	$CO_2 + 4\ H_2S + O_2 \rightarrow CH_2O + 4\ S + 3\ H_2O$	Symbiotic bacteria of tubeworms and clams	Vents and seeps
2	Aerobic methanotrophy	$CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$	Mussel symbionts; free-living bacteria	Seeps
3	Anaerobic oxidation of methane (AOM)	$CH_4 + SO_4^{2-} \rightarrow HCO_3^- + HS^- + H_2O$	Consortium of ANME archaea and sulfate-reducing bacteria	Oxygen-free sediment
4	Hydrogenotrophic methanogenesis	$CO_2 + 4\ H_2 \rightarrow CH_4 + 2\ H_2O$	Methanogenic archaea	Deep sediment
5	Acetogenesis	$2\ CO_2 + 4\ H_2 \rightarrow CH_3COOH + 2\ H_2O$	Acetogenic bacteria	Deep sediment
6	Nitrification, step 1	$NH_3 + 1.5\ O_2 \rightarrow NO_2^- + H^+ + H_2O$	Ammonia-oxidizing archaea (AOA)	The whole dark water column
7	Nitrification, step 2	$NO_2^- + 0.5\ O_2 \rightarrow NO_3^-$	Nitrite-oxidizing bacteria (NOB)	Same
8	Serpentinization (non-biological; simplified, iron end-member of olivine)	$3\ Fe_2SiO_4 + 2\ H_2O \rightarrow 2\ Fe_3O_4 + 3\ SiO_2 + 2\ H_2$	Water–rock reaction	Exposed mantle rock
9	Carbonate precipitation	$Ca^{2+} + 2\ HCO_3^- \rightarrow CaCO_3 + CO_2 + H_2O$	Chemical consequence of reaction 3	Seeps

Confidence in the equations: 0.9 **[C]** (standard formulations; 3 and 4 match those quoted in the sources consulted **[V]**).

6.3 Explanation of the terms used

- **Aerobic / anaerobic**: with / without oxygen (*aér* "air" + *bíos* "life"; *an-* "without").
- **Methanotroph**: "methane eater". **Methanogen**: "methane maker". They are opposite processes.
- **Archaea** (*archaîos* "ancient"): microorganisms without a nucleus, outwardly similar to bacteria but belonging to a different evolutionary lineage. All known methanogens are archaea.
- **ANME**: *ANaerobic MEthanotrophic archaea*, archaea that oxidize methane without oxygen.
- **Sulfate-reducing bacteria (SRB)**: bacteria that "breathe" sulfate instead of oxygen and turn it into sulfide.
- **Consortium / syntrophy** (*sýn* "together" + *trophé* "food"): two microbes that can only carry out a reaction by working together. In reaction 3, ANME oxidize the methane and pass the electrons to the SRB.
- **Hydrogenotrophic**: using hydrogen (H_2) as fuel.

6.4 How CO_2 is fixed: the pathways

To "fix" carbon is to convert CO_2 into organic molecules. There is no single way; at least six pathways are known. The relevant ones here **[C]** (0.85):

Pathway	Who uses it	Note
Calvin–Benson–Bassham cycle (CBB)	Sulfur-oxidizing symbionts; also plants	The same one photosynthesis uses
Reverse Krebs cycle (rTCA)	Many hydrothermal-vent bacteria	Cheaper in energy; oxygen-sensitive
Wood–Ljungdahl pathway (reductive acetyl-CoA)	Methanogens, acetogens and most SRB associated with ANME	Considered the most ancient; fixes CO_2 with H_2
3-hydroxypropionate / 4-hydroxybutyrate cycle	Ammonia-oxidizing archaea (Thaumarchaeota)	Very efficient at low energy

Verified finding: isotope experiments on seep sediments show that ANME/SRB consortia build their biomass mainly from **inorganic carbon**, not directly from methane, and that the SRB take up about eight times more inorganic carbon than the ANME. **[V]** (0.8). In other words, even the "methane-eating" microbes are, to a large extent, CO_2 fixers.

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7. The four kinds of chemical oasis

7.1 Hydrothermal vents

Hot springs on the seafloor, on ridges and submarine volcanoes. Seawater seeps down, is heated near magma, reacts with the rock and emerges loaded with H_2S and metals. **Black smokers** emit acidic fluid (pH 2–3) at several hundred degrees. They are rare in trenches, but they are where chemosynthetic ecosystems were first discovered.

7.2 Cold seeps

Places where methane- and sulfide-rich fluids ooze out of the sediment at ambient temperature. "Cold" only means they are not hot. **They are the dominant type in trenches**: compression of the accretionary prism squeezes the fluids out and faults lead them to the surface. **[V]** (0.9)

7.3 Serpentinization systems

Serpentinization: the reaction between seawater and **peridotite**, a mantle rock rich in the mineral **olivine**. It produces the mineral serpentine (a name alluding to its snakeskin look), heat, **hydrogen** and highly alkaline fluids. The hydrogen can react with inorganic carbon to form **abiotic** methane (made without life) and other small organic molecules such as formate.

Lost City (Atlantis Massif, about 15 km west of the Mid-Atlantic Ridge axis, at 750–900 m depth; discovered on 4 December 2000) **[V]** (0.95):

- Fluids from below 40 °C up to about 90 °C, with pH 9–11, rich in H_2 and CH_4.
- Carbonate chimneys 30 to 60 m tall; the largest, Poseidon, reaches about 60 m, making these the tallest vent structures known.
- Unlike black smokers, its fluids are poor in CO_2, H_2S and metals.
- Venting is very long-lived: published estimates range from at least 120,000 years to more than 150,000 years. (0.8)
- It needs no volcanic heat source: the reaction between mantle rock and seawater itself releases heat.
- A low diversity of methane-cycling archaea lives in the warm, porous interiors of the chimneys. The macrofauna is as diverse as at black-smoker sites on the Mid-Atlantic Ridge, but lacks their high biomass of chemosynthetic animals.

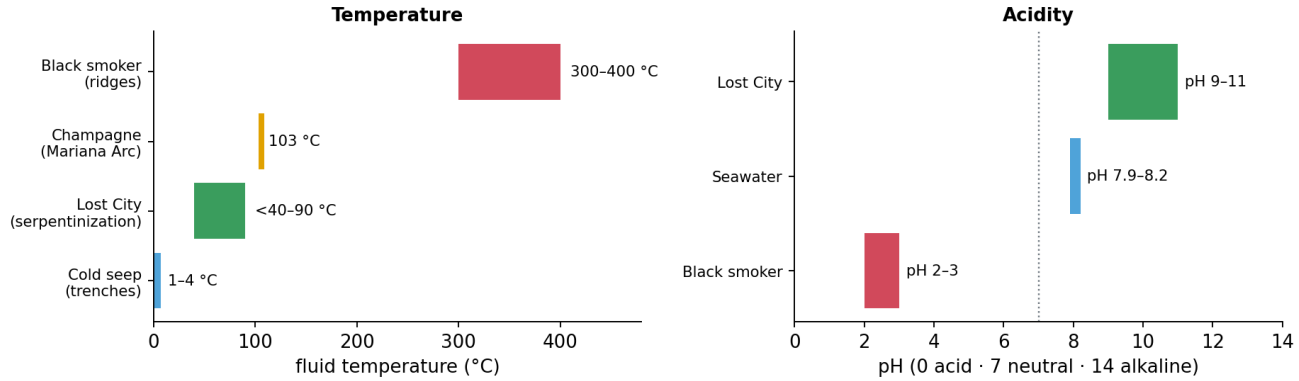
Shinkai Seep Field (southern Mariana forearc, discovered in 2010) **[V]** (0.9):

- At about 5,700 m, 80 km northeast of Challenger Deep.
- It is the deepest known serpentinite-hosted ecosystem.
- Dominated by the clam *Calyptogena (Abyssogena) mariana*, which lives on endosymbionts.
- Isotopes indicate that the community relies mostly on chemosynthetic production, although some organisms also use material of photosynthetic origin.

The Mariana forearc also hosts the only known series of active **serpentinite mud volcanoes** on the planet. **[V]** (0.85)

7.4 Organic falls

Whale carcasses and sunken logs. As they rot without oxygen they generate sulfide and attract chemosynthetic fauna for years or decades. They act as stepping stones between oases. **[C]** (0.8)



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Figure 9. Chemical oases differ greatly from one another: from 1 to 400 °C and from very acidic to very alkaline. Trench seeps are cold.

8. CO_2, the hidden protagonist

8.1 Basic chemistry of CO_2 in the sea

Dissolved CO_2 does not stay as CO_2. It reacts with water in an equilibrium of three forms:

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^- \rightleftharpoons 2\,H^+ + CO_3^{2-}$$

(dissolved gas) (carbonic acid) (bicarbonate) (carbonate)

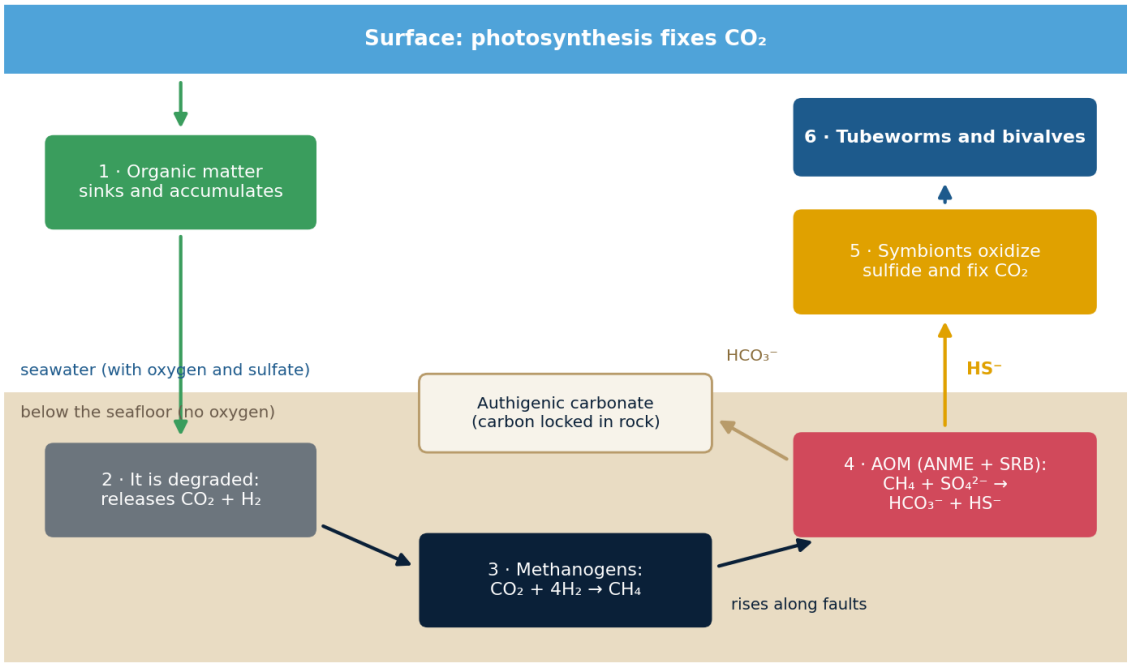
Together they are called **dissolved inorganic carbon (DIC)**. At seawater pH (~8) more than 90% is bicarbonate. When this document says "fixing CO_2", chemically it usually means DIC. **[C]** (0.9)

An important consequence: adding CO_2 releases H^+ and **acidifies**; consuming CO_2 or generating bicarbonate **raises alkalinity** and favors the precipitation of calcium carbonate ($CaCO_3$).

8.2 The three roles of CO_2

Role	What it means	Reaction in the table
Carbon source	The brick autotrophs build biomass with	1
Electron acceptor	Methanogens "breathe" it and turn it into CH_4	4 and 5
Product	Oxidizing methane returns the carbon to inorganic form	2 and 3

8.3 The carbon loop in a trench



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Figure 10. The carbon loop in a trench. CO_2 appears three times: released when organic matter is degraded, converted into methane, and fixed again by the symbiotic bacteria.

Step by step:

1. Organic matter (ancient photosynthesis) is buried along the trench axis.
2. Beneath the sediment, without oxygen, microbes degrade it and release CO_2 and H_2. This chemical transformation of buried sediment is called **diagenesis**.
3. Methanogens combine CO_2 and H_2 and produce methane.
4. The methane rises along faults. Part of it is trapped as **gas hydrate**.
5. Near the sediment surface, the ANME/SRB consortium oxidizes it with seawater sulfate. This process consumes about **90% of the methane** produced in marine sediments before it can escape. **[V]** (0.85)
6. Two products result: **sulfide**, which feeds the fauna's symbionts, and **bicarbonate**, which precipitates as **authigenic carbonate** (*authigenés* "born in place": a mineral formed on the spot, not transported).

8.4 How we know where the methane comes from: isotopes

An **isotope** (*ísos* "equal" + *tópos* "place": it occupies the same place in the periodic table) is a variant of an element with a different number of neutrons. Carbon has two stable ones: ^{12}C (light) and ^{13}C (heavy).

- Living things prefer ^{12}C. What they produce is "lightened".
- The ratio is expressed as **$\delta^{13}C$** (delta-13-C), in parts per thousand (‰). The more negative, the lighter.
- **Biogenic** or microbial methane: very negative $\delta^{13}C$ (typically below −60‰) and almost no other hydrocarbons. In seep sediments of the Laptev Sea, values of −98.9 to −77.6‰ were measured. **[V]**

- **Thermogenic** methane (cooked by heat at great depth): less negative and accompanied by ethane and propane.
- **Abiotic** methane (serpentinization): a different signature, with no biological origin.

In the 2025 study, methane made up 100% of the hydrocarbons measured and its isotopic signature indicates a microbial origin by CO_2 reduction. **[V]** (0.9)

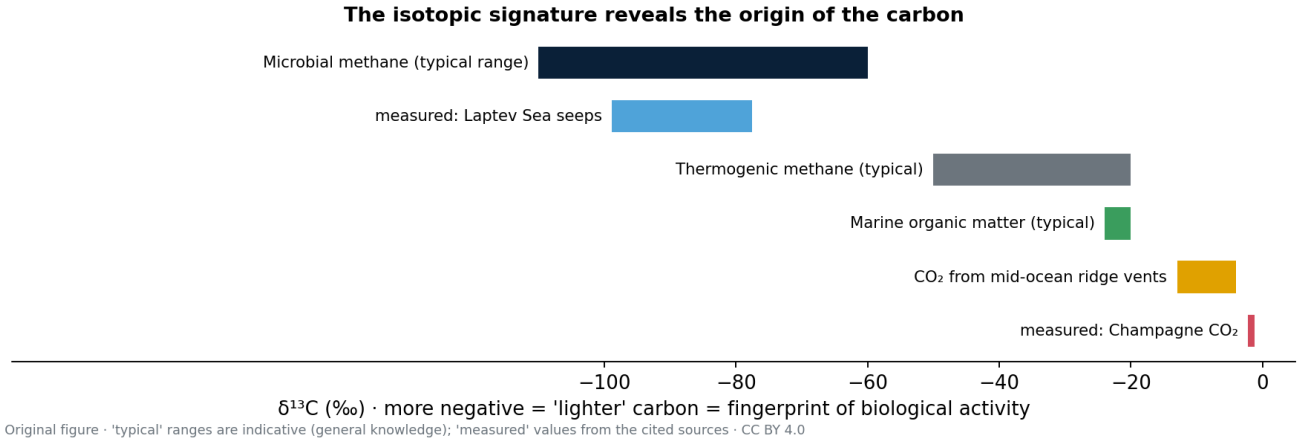


Figure 11. The $\delta^{13}C$ scale. Methane from hadal seeps falls in the microbial range, far to the left.

8.5 Liquid CO_2 on the seafloor

At high pressure and low temperature CO_2 is a liquid. The best-documented case is the **Champagne** site on the submarine volcano NW Eifuku (Mariana Arc) **[V]** (0.9):

- Depth: ~1,600 m.
- It discharges two fluids at once: a hot one (103 °C) with the highest CO_2 concentration ever reported in a submarine hydrothermal fluid (up to 2.7 mol/kg), and cold droplets (< 4 °C) of liquid CO_2.
- Composition of the droplets: ~98% CO_2, ~1% H_2S, traces of CH_4 and H_2.
- The droplets rise less than 200 m before dissolving. Liquid CO_2 floats because it is less dense than seawater above about 2,600 m; deeper down it would sink.
- It was the second place in the world where this was observed; the first was the Okinawa Trough (1990).
- There are mussel beds a few tens of meters away.
- The authors propose it as a natural laboratory for studying the effect of high CO_2 concentrations on marine life.

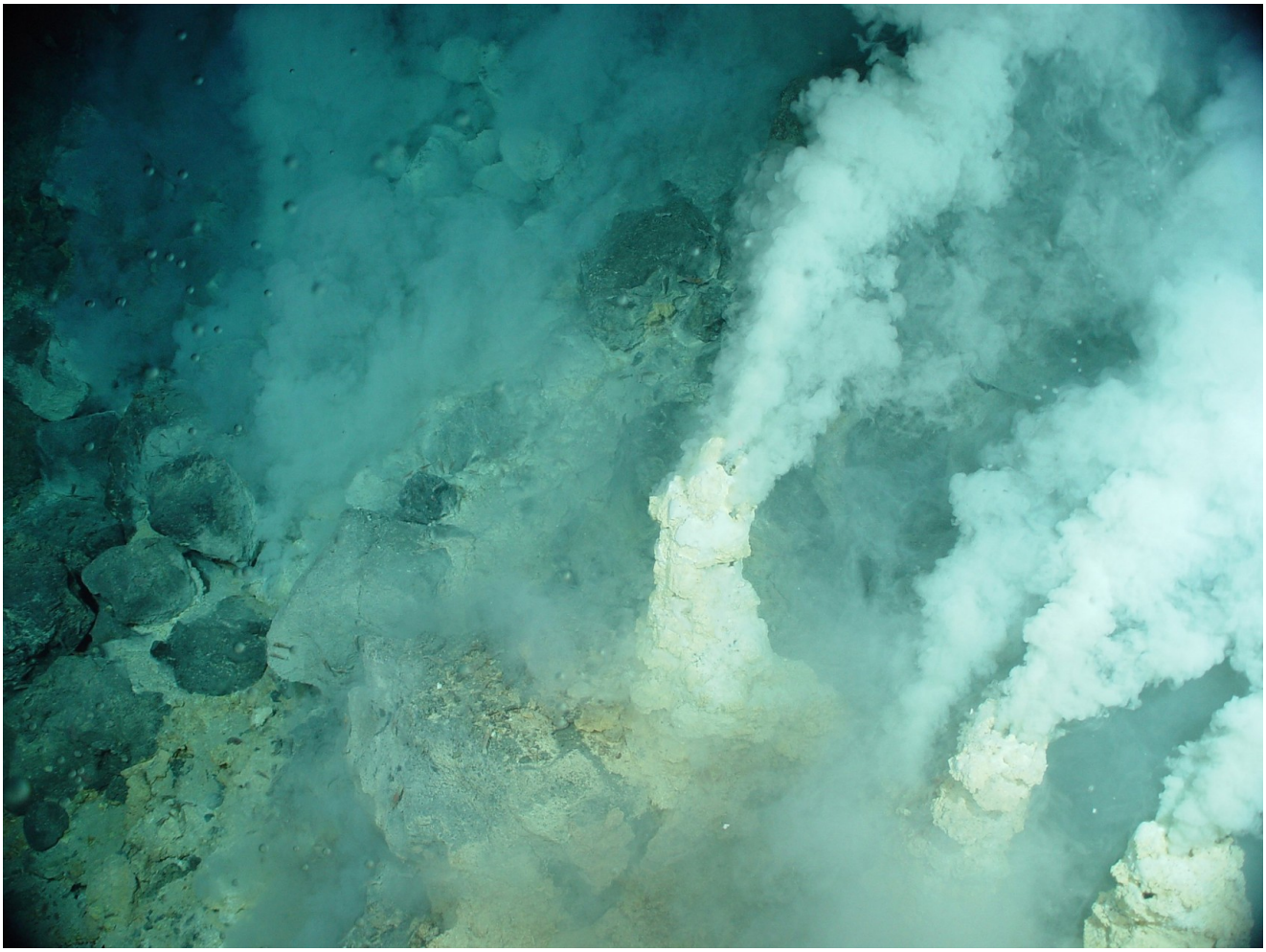


Photo 2. White chimneys at the Champagne site, about 20 cm across and 50 cm high, venting fluid at 103 °C. Droplets of liquid CO_2 can be seen rising on the left. Image: Pacific Ring of Fire 2004 Expedition, NOAA Office of Ocean Exploration; Dr. Bob Embley, NOAA PMEL, Chief Scientist. Public domain.

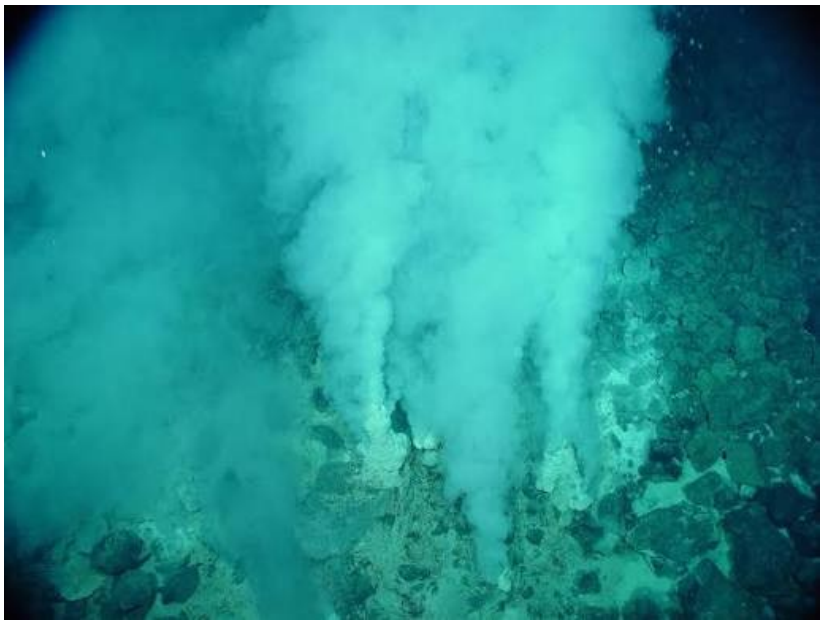


Photo 3. White smokers at the Champagne site seen from above. Image: Submarine Ring of Fire 2004 Exploration, NOAA Vents Program. Public domain.

Note: this is not a hadal trench but an arc volcano associated with the same subduction system. It is included because it shows the extreme end of CO_2 chemistry in that geological setting.

8.6 Gas hydrates

A **hydrate** or **clathrate** (Latin *clathratus*, "latticed") is an ice-like solid in which water molecules form cages that trap gas molecules (methane or CO_2). It is only stable when cold and under high pressure. Signs of hydrate decomposition were found in the hadal sediments in 2025, and layers of hydrate are proposed to exist beneath the seafloor. **[V]** (0.85)

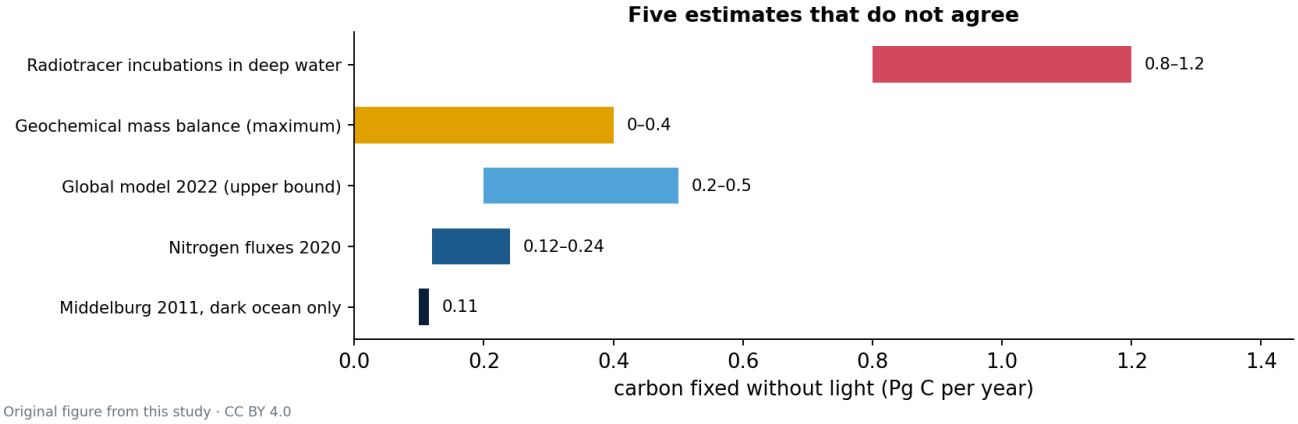
8.7 Dark carbon fixation outside the oases

Throughout the dark ocean, ammonia-oxidizing archaea (**Thaumarchaeota**) and nitrite-oxidizing bacteria fix CO_2 using the energy of **nitrification** (reactions 6 and 7).

The global estimates do not agree, and that is a result in itself **[V]** (0.7):

Estimate	Value
Radiotracer incubations in deep waters	0.8–1.2 Pg C/yr
Geochemical mass balances	up to 0.4 Pg C/yr
Global ecosystem model (2022), upper bound	0.2–0.5 Pg C/yr
Estimate based on nitrogen fluxes (2020)	1–2 × 10^{13} mol C/yr (≈ 0.12–0.24 Pg C/yr)
Total oceanic chemoautotrophy (Middelburg, 2011)	0.77 Pg C/yr, of which only 0.11 in the dark ocean

`Pg` = petagram = 10^{15} g = one billion metric tons. For reference, marine photosynthesis fixes on the order of 50 Pg C/yr **[C]**: chemosynthesis is small in quantity, but it is the only new production that exists in the dark.



Original figure from this study · CC BY 4.0

Figure 12. How much carbon is fixed without light in the ocean: published estimates differ by up to a factor of ten.

A counter-intuitive conclusion of that work: on a global scale, chemoautotrophy fueled by the recycling of organic matter is more important than that of hydrothermal vents. **[V]** (0.75)

8.8 The carbonate compensation depth

Below about 4,000–5,000 m (depending on the ocean), the water is so cold, compressed and CO_2-laden that calcium carbonate dissolves. That limit is called the **carbonate compensation depth (CCD)**. Trenches lie below it: sinking calcareous shells dissolve before they arrive, which constrains which shelled animals can live there and explains the interest of carbonates formed in situ by seeps. **[C]** (0.8)

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9. The 2025 discovery: chemosynthetic life at 9,533 m

Primary source: Peng, Du, Gebruk et al., *Flourishing chemosynthetic life at the greatest depths of hadal trenches*, **Nature 645, 679–685 (2025)**. Published online 30 July 2025. Open access.

9.1 Data

Item	Value	Confidence
Vehicle	*Fendouzhe* ("Striver"), a Chinese crewed submersible rated to full ocean depth	0.95 **[V]**
Lead institution	Institute of Deep-sea Science and Engineering (IDSSE), Chinese Academy of Sciences, with international collaboration	0.95 **[V]**
Trenches	Kuril–Kamchatka and western Aleutian (northwest Pacific)	0.95 **[V]**
Depth of the communities	5,800 – 9,533 m	0.95 **[V]**
Extent explored	~2,500 km	0.9 **[V]**
Cruise	8 July to 17 August 2024, aboard RV *Tansuoyihao*; 23 dives	0.9 **[V]**
First find	Dive FDZ 271: a dense community at 9,533 m, on black muds at the boundary between the trench bottom and the base of the accretionary prism	0.9 **[V]**
Dives with communities	19 of 23	0.8 **[V]** (secondary source)
Example of size	At the *Wintersweet Valley* field (9,120 m according to the text; 9,320 m according to the caption of Plate 2) the submersible followed the colonies for about 2 km, with thousands of worms	0.7 **[V]**
Geological setting	Base of the accretionary prism, where faults reach the seafloor	0.85 **[V]**
Fuel	Fluids rich in H_2S and CH_4	0.95 **[V]**
Origin of the methane	Microbial, by reduction of CO_2 derived from sedimentary organic matter	0.9 **[V]**
Previous record	Japan Trench: 7,326 m (1999), later extended to 7,434 m	0.85 **[V]**

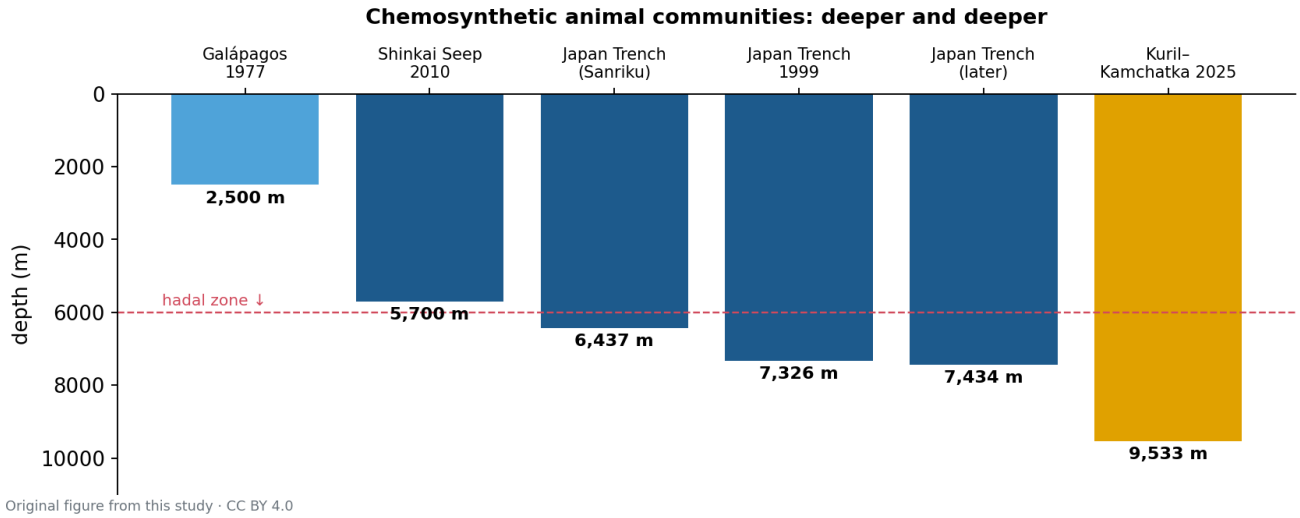


Figure 13. Depth of the known chemosynthetic animal communities. The 2025 discovery exceeds the previous record by more than 2,000 m.

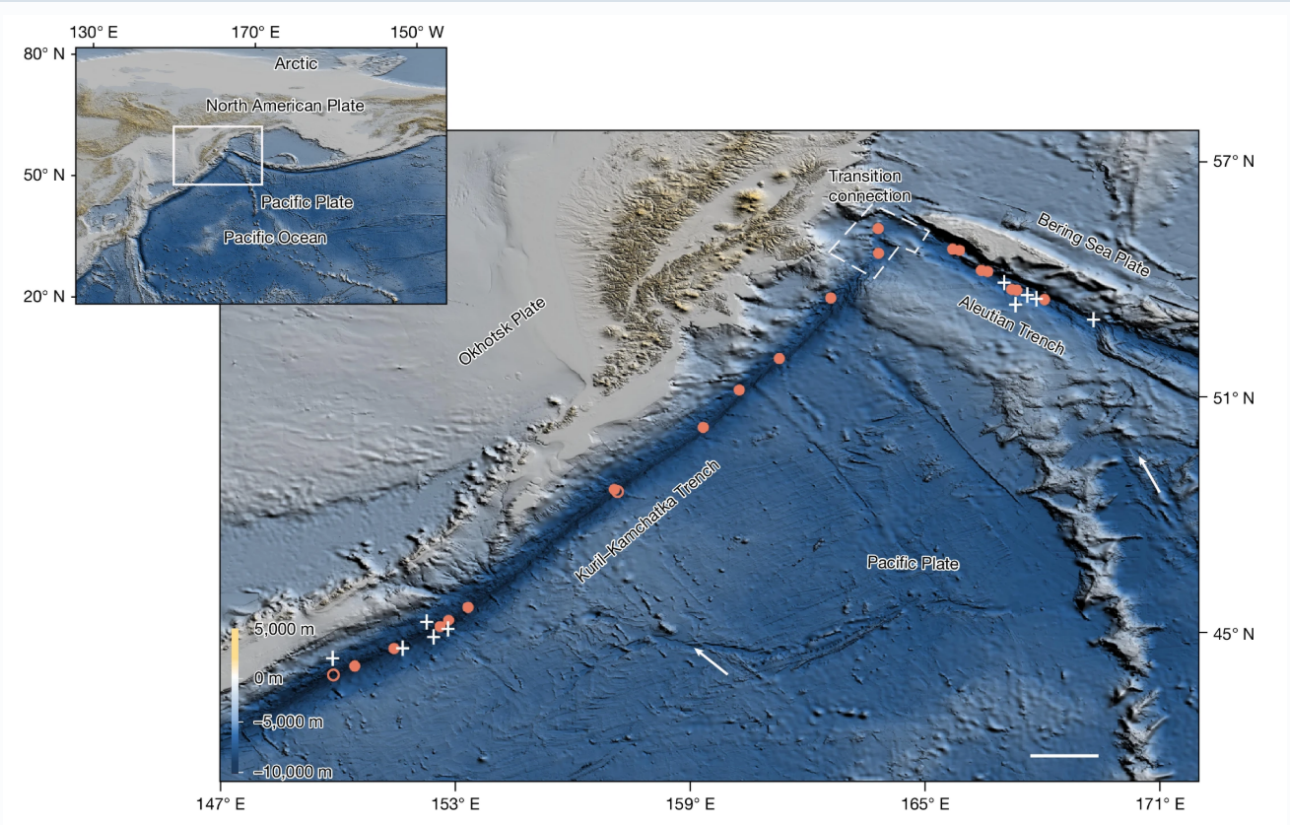


Plate 1. Map of the study area: Kuril–Kamchatka Trench, transition zone and Aleutian Trench, with the tectonic plates involved. Orange dots: dives on which chemosynthetic communities were observed and sampled. Crosses: dives without them. Open orange circles: potential seep sites with black sediments. White arrows: direction of subduction. White dashed line: transition zone between the two trenches. Reproduced without modification from Peng et al., *Nature* 645, 679–685 (2025), license CC BY-NC-ND 4.0.

9.2 Fauna

- **Kuril–Kamchatka** (from ~7,000 to 9,533 m): dominated by **frenulate siboglinids**.
- **Western Aleutian and transition zone** (generally shallower than 7,000 m): dominated by **bivalves**, siboglinids or tube-dwelling polychaetes of the family Ampharetidae, depending on the site.

- The species composition of siboglinids varies considerably from one community to another within the same trench.

Terms:

- **Siboglinids**: a family of annelid worms with no mouth or gut as adults, living in tubes and wholly dependent on symbiotic bacteria. *Riftia* is the best-known member.
- **Frenulates**: a subgroup of very thin siboglinids (often under 1 mm across) that live buried in sediment. The name comes from the *frenulum*, a small ridge ("little bridle" in Latin) on the body.
- **Polychaetes** (*polýs* "many" + *chaíte* "bristle"): bristled marine annelid worms.
- **Bivalves**: mollusks with a two-part shell (clams, mussels).

The places, by name. The authors named the fields they found. Plate 2 shows eight images of them, and the table that follows explains each one.

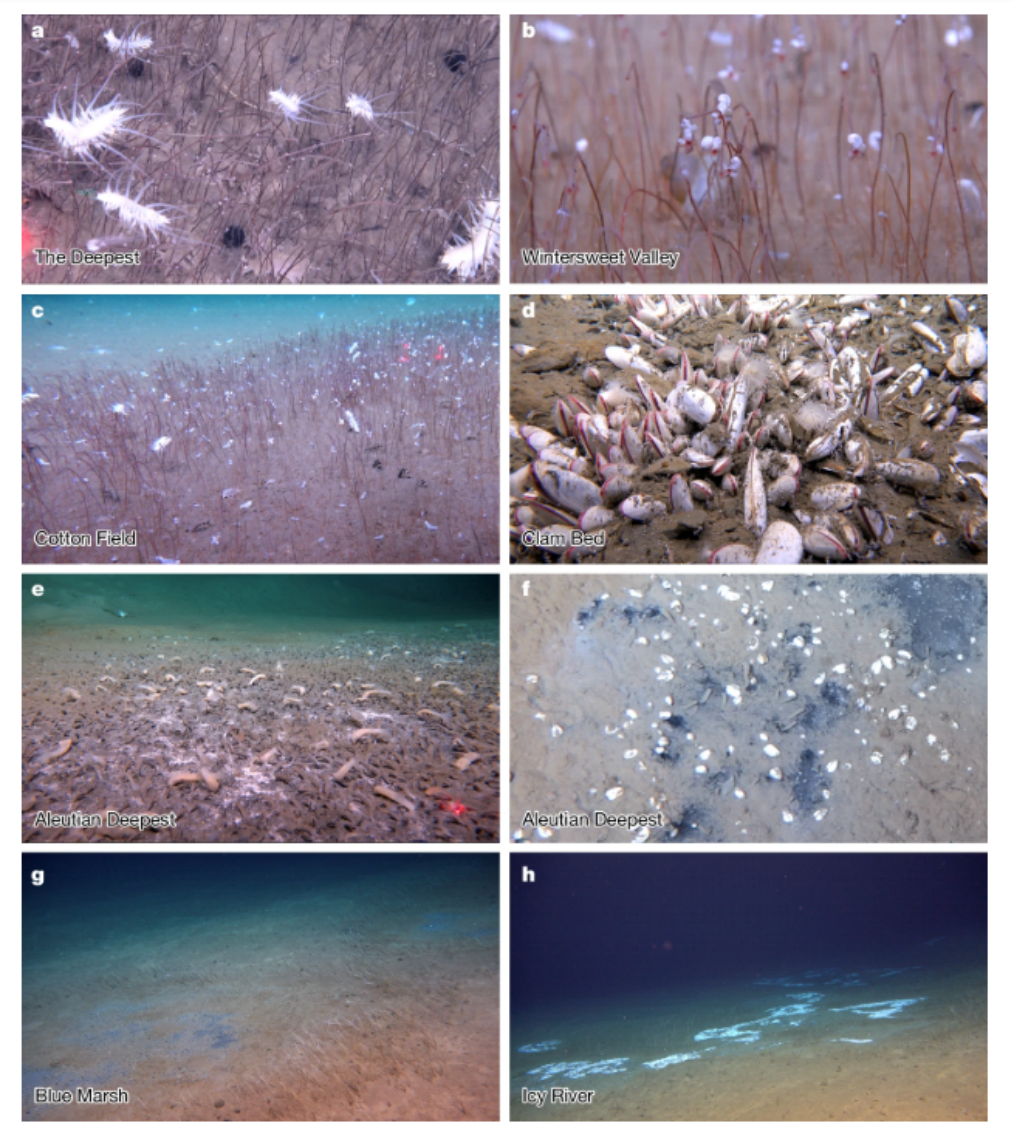


Plate 2. The chemosynthetic communities found, with the names given by their discoverers. Reproduced without modification from Peng et al., *Nature* 645, 679–685 (2025), license CC BY-NC-ND 4.0.

Panel	Name and depth	What it shows, according to the original figure caption
a	*The Deepest*, 9,532 m	Free-moving polychaetes *Macellicephaloides grandicirra* (white, up to 6.5 cm) among dense colonies of frenulate siboglinids with tubes 20–30 cm long and about 1 mm in diameter

Panel	Name and depth	What it shows, according to the original figure caption
b	*Wintersweet Valley*, 9,320 m	Clusters of frenulates extending red, hemoglobin-filled tentacles. The white spots on the tubes, near the tentacles, are small gastropods
c	*Cotton Field*, 9,332 m	Tightly packed frenulates hosting abundant free-moving *M. grandicirra* (white)
d	*Clam Bed*, 5,743 m	Dense aggregation of vesicomyid bivalves *A. phaseoliformis* (up to 23 cm) in the sediment, with about 6–8 cm of shell exposed and often carrying sea anemones (Actiniaria)
e	*Aleutian Deepest*, 6,870 m	Tube-dwelling polychaetes *Anobothrus* sp. and sea anemones dominate, with spots of white microbial mats
f	*Aleutian Deepest*, 6,928 m	Dense aggregation of vesicomyid bivalves *I. fossajaponicum* (up to 3 cm) associated with black sediments, with tube-dwelling *Anobothrus* sp.
g	*Blue Marsh*, 6,800 m	Dark blue muds surrounded by clusters of frenulates, marking methane seeps
h	*Icy River*, 6,700 m	Large patches of white, snow-like microbial mats stretching tens of meters, accompanied by frenulates

The whole table follows the original figure caption **[V]** (0.95). The images were taken with the submersible's high-definition camera; the distance between the laser points is 10 cm, which gives the scale.

Two depths that the paper itself gives in two ways. Both versions were checked in the text of the paper **[V]** (0.9):

Field	Original caption of Plate 2	Text of the paper
Wintersweet Valley	9,320 m (panel b)	"at a depth of 9,120 m (dive FDZ 274)"
Cotton Field	9,332 m (panel c)	*Dead Valley* and *Cotton Field*, "at depths of 9,522 and 9,566 m" (dive FDZ 275); they extend at least 2.2 km along the 9,520 m contour

The difference is in the source. It may be because these are kilometer-long fields on a sloping bottom, so the photograph and the reference point need not be at the same depth, or it may be a slip in the paper; this cannot be settled without the dive table in the supplementary information. This report uses the caption figure when discussing the photographs and the figure from the text when discussing the field.

9.3 Geochemistry and geological model

Where the methane comes from. The classic diagram for identifying the origin of methane plots two isotopes against each other: carbon ($\delta^{13}C$) and hydrogen (δD, the proportion of deuterium). The trench samples (Plate 3, panel a) fall at $\delta^{13}C$ values between about −80 and −97‰, inside the "CR" field, which corresponds to **carbonate reduction**, that is, methanogenesis from CO_2 (reaction 4). They lie far from the thermogenic and abiotic fields. **[V]** (0.9; values read from the figure)

Why there are hydrates. Panel b of Plate 3 is a phase diagram: it shows what form methane takes depending on pressure and temperature. The samples, taken at about 275 K (2 °C) and between about 58 and 98 MPa, fall in the "H + L" zone: **hydrate plus liquid**. Under those conditions, methane that saturates the water forms stable solid hydrate. **[V]** (0.85)

How the system works. Plate 4 summarizes the authors' model:

1. Organic matter arrives down the slope and accumulates in the sediment.
2. The sinking plate drags that sediment beneath the **accretionary wedge**, along the **décollement** (the detachment surface between the two plates).
3. Below the seafloor, microbes generate methane, part of which remains as lenses of **hydrate** between the layers.
4. Methane-rich fluids migrate along the layers and rise along **normal faults** in the subducting plate.
5. Where the fault reaches the seafloor the seep appears, with its community of tubeworms and clams and with deposits of **ikaite**.

Ikaite is a hydrated calcium carbonate ($CaCO_3 \cdot 6H_2O$) that is stable only in very cold water; at room temperature it breaks down. Its presence is another sign that inorganic carbon is precipitating there. **[C]** (0.85)

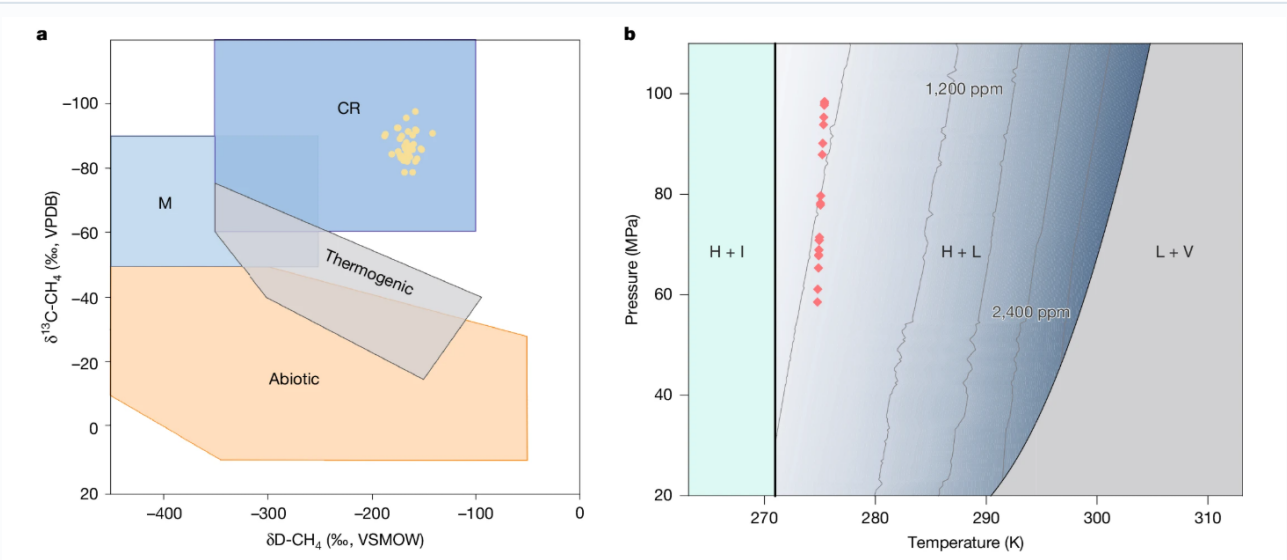


Plate 3. Panel a: origin of the methane according to its isotopes; the yellow dots are the samples. Panel b: stability of methane hydrate; the red diamonds are the samples. Reproduced without modification from Peng et al., *Nature* 645, 679–685 (2025), license CC BY-NC-ND 4.0.

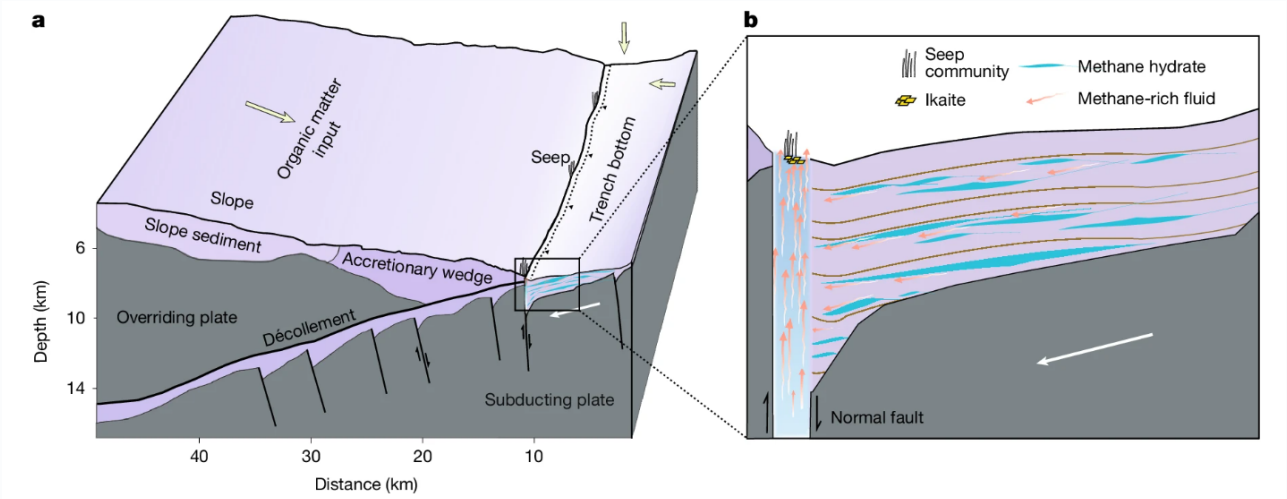


Plate 4. Geological model for the formation of hadal cold seeps. Light green arrows: input of organic matter, downward and lateral. White arrow: direction of subduction. Dashed line: trench axis, nearly parallel to the band of seeps. Panel b enlarges the base of the accretionary wedge. Reproduced without modification from Peng et al., *Nature* 645, 679–685 (2025), license CC BY-NC-ND 4.0.

9.4 Why it matters

1. **It changes the model of hadal feeding.** Trench fauna was assumed to live on what sinks. Here a substantial part lives on local chemical energy.
2. **It implies an active microbial biosphere beneath the seafloor.** That much microbial methane means very active methanogen communities in deep hadal sediments.
3. **It affects the carbon cycle.** Trenches may store and recycle more carbon than models account for.
4. **The "chemosynthetic corridor" hypothesis.** The authors propose that, since many trenches share this geology, such communities could be common worldwide.

9.5 Caveats

- Some media wrote "9,533 kilometers". That is a typo: it is **meters**.
- It is not the deepest hint of chemosynthetic microbes: a 2012 expedition photographed possible microbial mats at about 10,700 m in the Mariana Trench (published in 2020), but with no documented animal community. The 2025 record is for an **animal community** sustained by chemosynthesis. **[V]** (0.7, secondary source)
- "Thousands of worms" is supported for particular fields. Extrapolating to totals for the whole trench is not.
- Animals found deeper still, such as the amphipods at nearly 11,000 m, are not chemosynthetic.

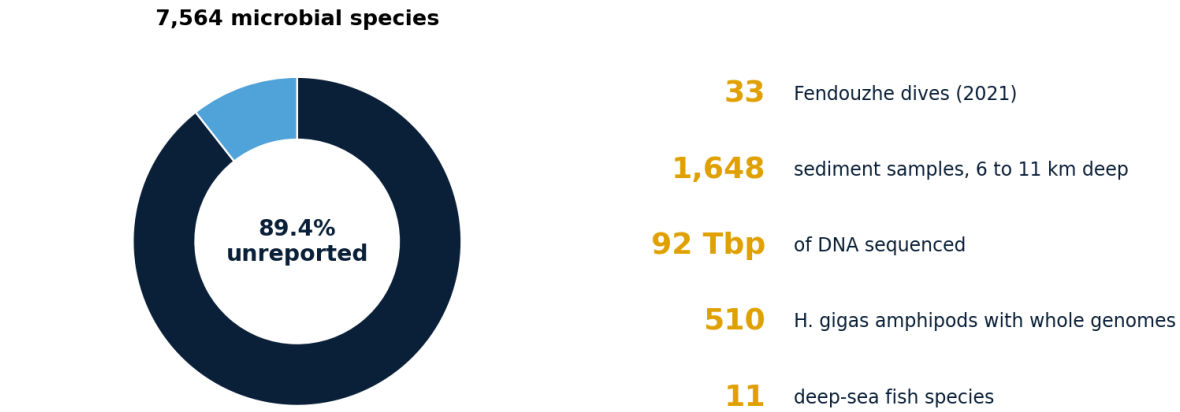
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10. Who lives down there

10.1 The microbes: the MEER project

MEER (*Mariana Trench Environment and Ecology Research*) is the first systematic study of the hadal ecosystem. Its first results were published in the journal *Cell* on 6 March 2025 (one commentary and three research papers). **[V]** (0.9)

Item	Value
Fendouzhe dives (2021)	33, in the Mariana Trench, the Yap Trench and the Philippine Basin
Sediment samples	1,648, between 6 and 11 km depth
Volume of genetic data	92 Tbp (terabase pairs: 92 trillion "letters" of DNA)
Microbial species identified	7,564
Previously unreported	89.4%
What drives that diversity	Homogeneous selection (50.5%) and dispersal limitation (43.8%); random drift plays a minimal role



Original figure with data from the MEER project (Cell, 2025) · CC BY 4.0

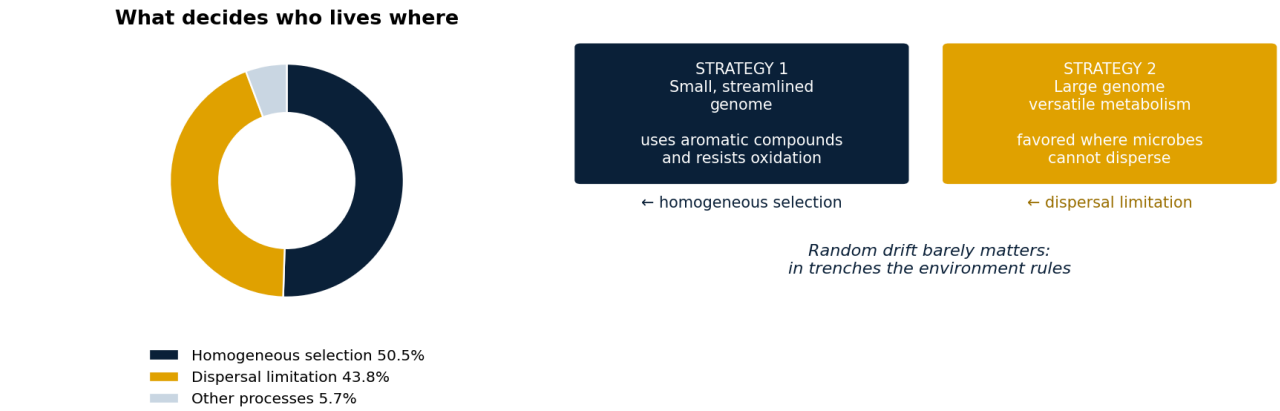
Figure 14. The MEER project in numbers.

Two microbial strategies detected **[V]** (0.8):

- Small, "streamlined" genomes with key functions for using aromatic compounds (adaptation to scarcity) and for antioxidation (adaptation to pressure).
- Large genomes with versatile metabolism, favored where dispersal is limited.

Terms:

- **Metagenomics** (*metá* "beyond"): sequencing all the DNA present in an environmental sample, without culturing the microbes one by one.
- **16S rRNA**: a gene present in all bacteria and archaea, used as a "barcode" to identify them.
- **Prokaryote** (*pró* "before" + *káryon* "kernel, nucleus"): a cell without a nucleus. Includes bacteria and archaea.
- **Homogeneous selection**: similar environmental conditions selecting the same kinds of microbes at different sites.
- **Dispersal limitation**: microbes cannot move freely between sites, so each place develops its own community.



Original figure with data from the MEER project (Cell, 2025) · CC BY 4.0

Figure 20. Left: processes that determine which microbes live at each point in the trench. Right: the two genomic strategies detected.

10.2 Amphipods: the masters of the bottom

Amphipods (*amphí* "on both sides" + *poús* "foot") are crustaceans that look like small shrimp. They are the dominant scavengers of the hadal zone.

Hirondellea gigas **[V]** (0.9):

- It lives between about 6,800 and 11,000 m, including Challenger Deep.
- A baited trap lowered for three hours into Challenger Deep caught almost 200 specimens.
- It produces enzymes that digest plant polysaccharides: amylase, **cellulase**, mannanase and xylanase. Its cellulase converts cellulose into glucose and cellobiose, and obtains glucose directly from dried wood.
- The enzyme's activity **increases** at 100 MPa and 2 °C, the conditions of Challenger Deep.
- Unlike termites, it does not seem to depend on gut microbes to digest wood.
- The authors' conclusion: it adapted to a poor environment by eating sunken plant debris.
- Its genome is enormous: 13.92 Gb, more than four times the human genome. Sequencing 510 individuals revealed no genetic differences between populations at different depths, suggesting it tolerates a very wide pressure range.
- A 2019 paper proposes that it coats its body with an aluminum gel extracted from the sediment, which would protect it. (0.7; only the title of the study was verified)

An **enzyme** is a protein that speeds up a specific chemical reaction. **Cellulase** breaks down cellulose, the main component of wood; almost no animal makes it by itself.

Alicella gigantea: a giant amphipod up to about 30 cm long, an example of **deep-sea gigantism** (the tendency of some deep invertebrates to be far larger than their shallow-water relatives). **[C]** (0.8)

Eurythenes plasticus: see 12.4.

10.3 Fish: the 8,400 m limit

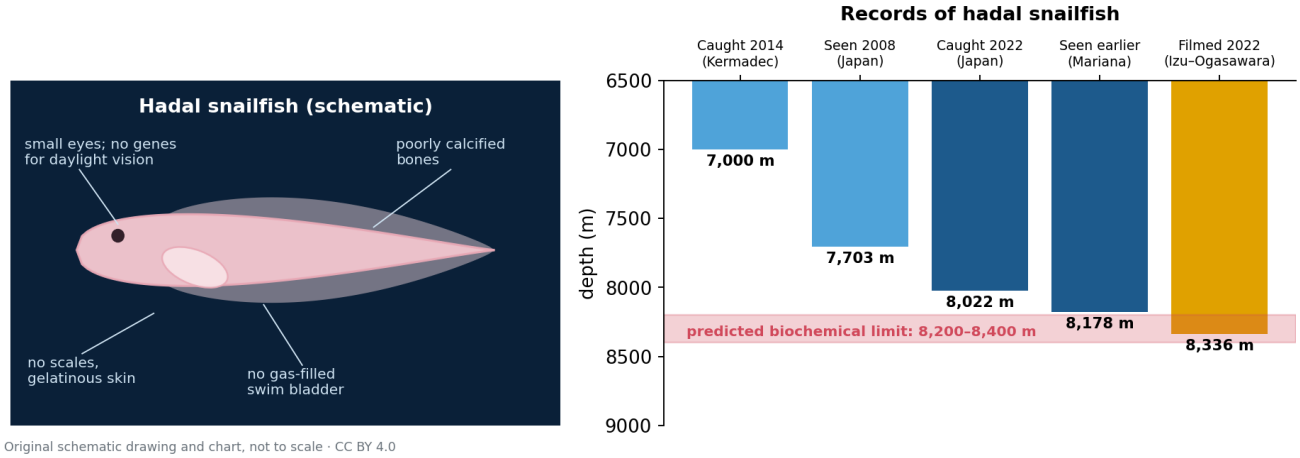
Snailfish (family Liparidae): scaleless, gelatinous-bodied, with more than 400 species in about 30 genera, from the shoreline to the trenches. They have the widest depth range of any fish family. **[V]** (0.9)

Record	Depth	Detail
Deepest fish ever filmed	**8,336 m**	*Pseudoliparis* sp., a juvenile, Izu–Ogasawara Trench, 15 August 2022. Baited camera on an autonomous lander
Deepest fish ever caught	**8,022 m**	Two *Pseudoliparis belyaevi*, Japan Trench, days later. First fish collected from below 8,000 m
Previous record (observation)	8,178 m	Mariana Trench

[V] (0.9). The expedition was run by the University of Western Australia and the Tokyo University of Marine Science and Technology. The lead scientist had predicted ten years earlier that the limit would lie between 8,200 and 8,400 m and that it would be a snailfish.

Genomic data from MEER: hadal snailfish have completely lost the genes for daylight (**photopic**) vision and have enhanced those for low-light vision. **[V]** (0.8)

No fish has been found in the deepest 25% of the ocean (8,400–11,000 m). The explanation is in section 11.



Original schematic drawing and chart, not to scale · CC BY 4.0

Figure 21. Left: features of a hadal snailfish (schematic drawing). Right: depth records and the predicted biochemical limit.

Profile of one species: the Mariana snailfish (*Pseudoliparis swirei*)

It is the best-known hadal fish. The data come from the paper that described it (Gerringer et al., 2017), read in full. **[V]** (0.95)

Aspect	Data
Description	2017, from 37 specimens caught in mackerel-baited traps on two cruises (RV *Falkor*, 2014, and RV *Shinyo-maru*, 2017)
Depth	Caught between 6,898 and 7,966 m; filmed between 6,198 and 8,098 m. Abundant between 7,000 and 8,000 m. None was seen below 8,200 m, which is precisely the physiological limit proposed for fish
Size	89 to 235 mm standard length
Appearance in life	Pinkish-white body. The skin is transparent: the liver, stomach and muscles can be seen through it
Eyes	Very small, about 10% of head length
Skeleton	Bones fully developed but weakly calcified
Teeth	Simple, sharp canines in bands; well-developed pharyngeal teeth

Aspect	Data
Feeding	Gathers in groups at bait and feeds on the amphipods that swarm to it
Reproduction	Eggs up to 9.4 mm in diameter, among the largest known in bony fish. Each female carries a few large eggs (up to 23) and hundreds of small ones
Size and depth	Smaller individuals were caught deeper
Relationships	Genetically closer to *P. belyaevi* of the Japan Trench than to the Kermadec snailfish. The authors question whether the genera *Pseudoliparis* and *Notoliparis* are really distinct
Distribution	Almost certainly endemic to the Mariana Trench
Name	In honor of Herbert Swire, navigating officer of HMS *Challenger*, the expedition that discovered the trench in 1875

At the time it was the deepest capture of a vertebrate with corroborated depth data; it was surpassed by the specimens from 8,022 m in the Japan Trench (2022).

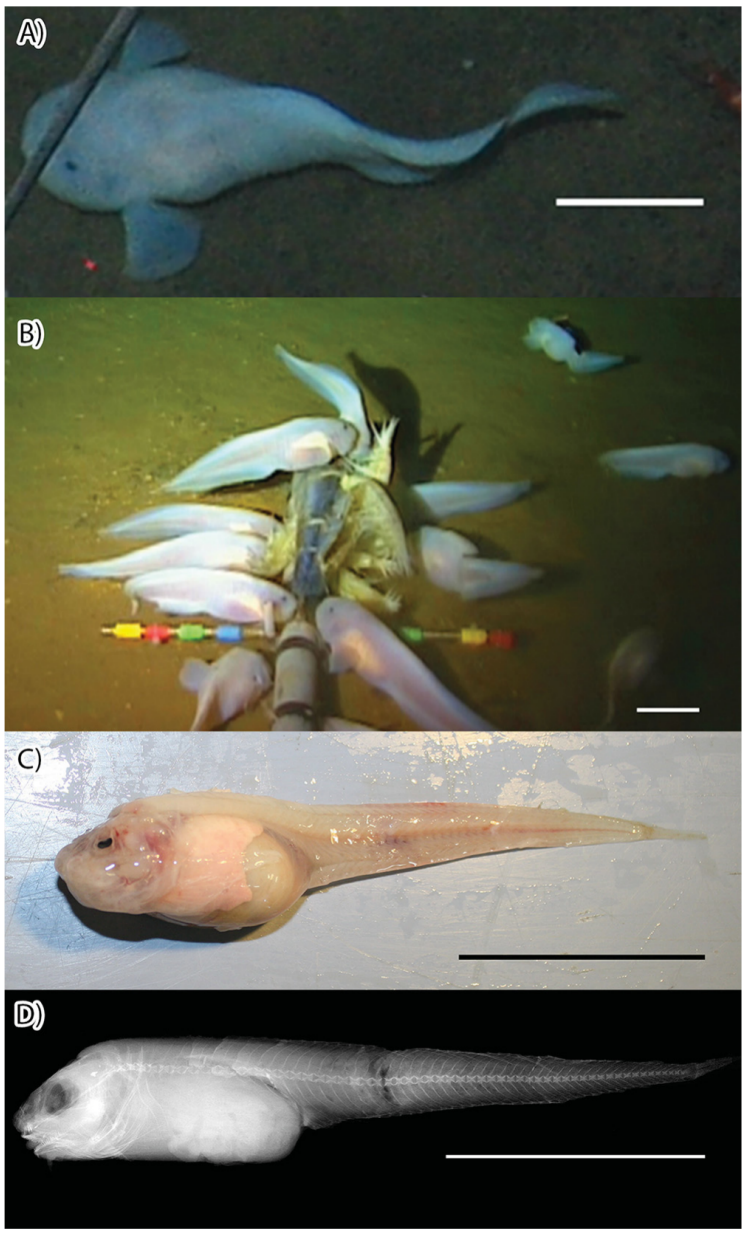


Plate 5. *Pseudoliparis swirei*. A) Photographed on the seafloor at 6,198 m. B) A group at 7,485 m, around the bait. C) Freshly caught specimen, on deck. D) Radiograph of another specimen (image by Sandra Raredon). Scale bars: 5 cm. Reproduced from Gerringer et al., *Zootaxa* 4358, 161–177 (2017), license CC BY 3.0.

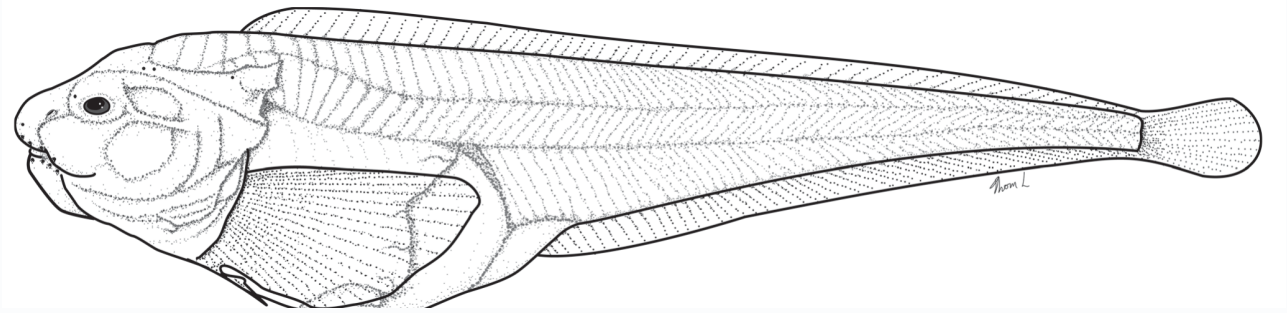


Plate 6. Lateral view of *Pseudoliparis swirei*, scientific drawing by Thomas D. Linley from the holotype, the paratypes and photographs of a freshly caught 151 mm juvenile. Reproduced from Gerringer et al., *Zootaxa* 4358, 161–177 (2017), license CC BY 3.0. Cropped from the original page.

10.4 Other inhabitants

Organism	Fact	Confidence
Holothurians (sea cucumbers), mainly the genus *Elpidia* and relatives	Dominate the visible biomass on many hadal floors; they eat sediment. About 59 hadal species	0.85
Brittle stars	Present in several trenches; *Ophiolimna* sp. in the Kermadec Trench	0.8
Xenophyophores	**Single-celled** organisms reaching several centimeters; observed below 10,000 m. The name means "bearer of foreign bodies": they build their framework by gluing particles together	0.8
Foraminifera	Single-celled shelled protists; very abundant in Challenger Deep sediment	0.8
Polychaetes, isopods, gastropods, tanaids, nematodes, sea anemones	Present down to the greatest depths	0.8

Decapod crustaceans (shrimp, crabs) disappear at around 7,000–8,000 m. (0.7)

Further data on the floor of the Mariana Trench **[V]** (0.8):

- In the water of Challenger Deep, microbial biodiversity **increases** with depth: a 2020 study reconstructed 726 genomes from 27 phyla between the surface and 10,500 m.
- The largest holothurians observed by the 2012 expedition were the biggest animals on the trench floor, and giant amphipods 17 cm long were caught at nearly 11 km.

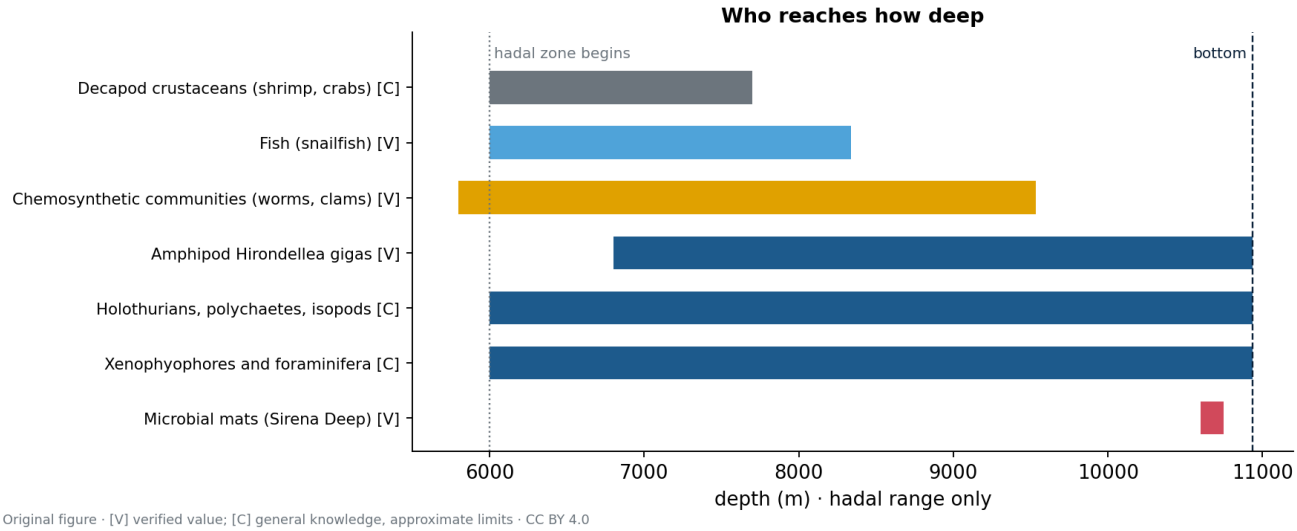
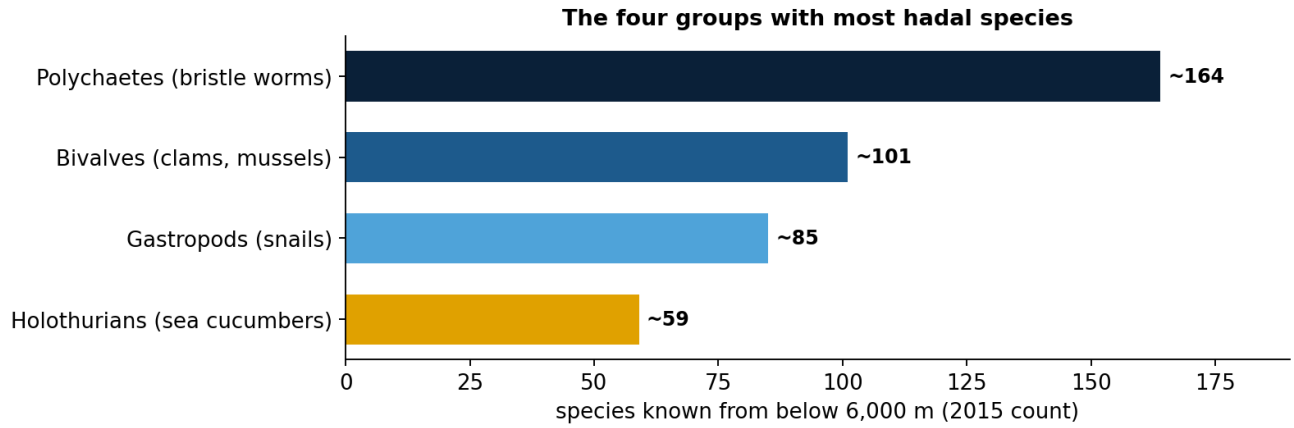


Figure 22. How deep each group reaches within the hadal zone. Only data marked [V] come from checked sources.

10.5 Catalog of hadal species

It is not possible to list every species: in four groups alone, about 400 were known from below 6,000 m in 2015, and every cruise adds more. **[V]** (0.85)



Original figure with data from Jamieson (2015), The Hadal Zone · CC BY 4.0

Figure 23. Species known from the hadal zone in the four most diverse groups, according to Jamieson's (2015) count. All of them reach full ocean depth.

This catalog gathers the best-documented species in the sources consulted. The last column shows where each appears in the images of this report.

Fish

Species	Where	Depth	Notable fact	Image in this report
Pseudoliparis swirei, Mariana snailfish	Mariana Trench	Caught at 6,898–7,966 m; filmed at 6,198–8,098 m (FishBase gives down to 8,178 m)	See the profile in section 10.3	Plates 5 and 6
"Ethereal snailfish" (no scientific name)	Mariana Trench	7,939–8,145 m	Only filmed, never caught	—
Pseudoliparis belyaevi	Japan Trench	Caught at 8,022 m	First fish collected from below 8,000 m (2022)	—
Pseudoliparis sp. (juvenile)	Izu–Ogasawara Trench	8,336 m	Deepest fish ever filmed	—
Notoliparis kermadecensis	Kermadec Trench	Caught at 7,000 m; seen at 7,500 m	Used to measure TMAO (section 11.3)	—
Paraliparis selti	Atacama Trench	Hadal	Endemic to that trench	—

Each trench has its own snailfish: hadal species appear isolated to a single trench or trench system (Kermadec, Macquarie, South Sandwich, South Orkney, Peru–Chile, Kuril–Kamchatka, Japan). Other fish families barely touch the hadal zone: cutthroat eels at 6,068 m, eelpouts at 6,145 m and grenadiers down to 7,012 m. **[V]** (0.85)

Amphipods

Species	Where	Depth	Notable fact	Image in this report
Hirondellea gigas	Mariana and other northwest Pacific trenches	~6,800–11,000 m	Digests wood with its own cellulase; 13.92 Gb genome	—

Species	Where	Depth	Notable fact	Image in this report
Hirondellea thurstoni	Atacama Trench	Hadal	Endemic to that trench	—
Alicella gigantea, supergiant amphipod	Kermadec, Mariana	Abyssal and hadal	The largest known amphipod; at the top of the trench food web	—
Eurythenes plasticus	Mariana Trench	6,010–6,949 m	Described in 2020 with a plastic fiber in its gut	—
Eurythenes atacamensis	Atacama Trench	Hadal	Endemic to that trench	—
Dulcibella camanchaca	Atacama Trench	7,902 m	Described in 2024; a new genus. The first large **predatory** amphipod of the hadal zone (almost 4 cm). Its name alludes to darkness. Its predatory habit is inferred from its anatomy; it has not been seen hunting	—

Other crustaceans and cephalopods

Species	Where	Depth	Notable fact	Image in this report
Benthesicymus crenatus, prawn	Japan Trench and many others	Down to 7,703 m	The deepest decapod observed. Comes to bait to prey on amphipods	—
Acanthephyra sp., prawn	Kermadec Trench	Down to 6,890 m	Seen passing, with no interest in the bait	—
Grimpoteuthis sp., Dumbo octopus	Java Trench (Indian Ocean)	6,957 m	The deepest octopus; first firm evidence of a hadal cephalopod (observed in 2019)	—
Magnapinna sp., bigfin squid	Philippine Trench	6,212 m	The deepest squid; first hadal squid (published in 2021)	—

Chemosynthetic fauna

Species	Where	Depth	Notable fact	Image in this report
Frenulate siboglinids of the genera *Lamellisabella* and *Polybrachia*, among others	Kuril–Kamchatka	~7,000–9,533 m	Dominate the 2025 communities; tubes 20–30 cm long	Plate 2, panels a, b and c
Macellicephaloides grandicirra, free-moving polychaete	Kuril–Kamchatka (*The Deepest*, *Cotton Field*)	9,332–9,532 m	Moves among the frenulate tubes; up to 6.5 cm. Not chemosynthetic itself: it lives in the community	Plate 2, panels a and c
Maorithyas hadalis, clam (reclassified in 2024 as *Tartarothyas hadalis*)	Japan Trench	7,326 m	Two kinds of bacteria in its gill	—
Calyptogena (Abyssogena) phaseoliformis, clam	Japan Trench; Aleutian Trench (*Clam Bed*)	6,437 m; 5,743 m	Held the chemosynthetic depth record before 1999. Up to 23 cm	Plate 2, panel d

Species	Where	Depth	Notable fact	Image in this report
Spirobrachia sp., frenulate	Kuril–Kamchatka (*The Deepest*)	~9,530 m	Collected at the deepest field	—
I. fossajaponicum, vesicomyid clam	Aleutian (*Aleutian Deepest*)	6,928 m	Up to 3 cm; forms dense aggregations on black sediment. The caption abbreviates the genus	Plate 2, panel f
Anobothrus sp., tube-dwelling polychaete (ampharetid)	Aleutian (*Aleutian Deepest*)	6,870–6,928 m	Dominant together with sea anemones	Plate 2, panels e and f
Small gastropods (unidentified)	Kuril–Kamchatka (*Wintersweet Valley*)	9,320 m	Live on the frenulate tubes, near the tentacles	Plate 2, panel b
Calyptogena (Abyssogena) mariana, clam	Shinkai Seep Field	5,500–5,700 m	Lives on serpentinite	—
Riftia pachyptila, giant tubeworm	East Pacific ridges	~2,500 m (not hadal)	The species in which the symbiosis was discovered	Photo 1

Groups for which no particular species is highlighted (holothurians, brittle stars, single-celled organisms) are in section 10.4. Overall confidence of the tables: 0.85 for place and depth, which come from the cited sources.

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11. Adaptations to pressure

11.1 The problem

Pressure does not "crush" an animal that has no air spaces, because water is almost incompressible. The damage is molecular: pressure forces water molecules into **proteins**, deforming them so that they stop working, and it stiffens cell membranes.

An organism adapted to high pressure is a **piezophile** (*piézein* "to press" + *phílos* "friend").

11.2 The solution: piezolytes

An **osmolyte** is a small molecule that cells accumulate to regulate their water content. Some also protect proteins from pressure: these are **piezolytes** ("pressure solutes").

The main one in fish is **TMAO** (trimethylamine N-oxide). It acts as a "chemical chaperone", preventing water from entering proteins. When it breaks down it produces trimethylamine, responsible for the characteristic "fishy" smell.

11.3 Why fish do not go below 8,200–8,400 m

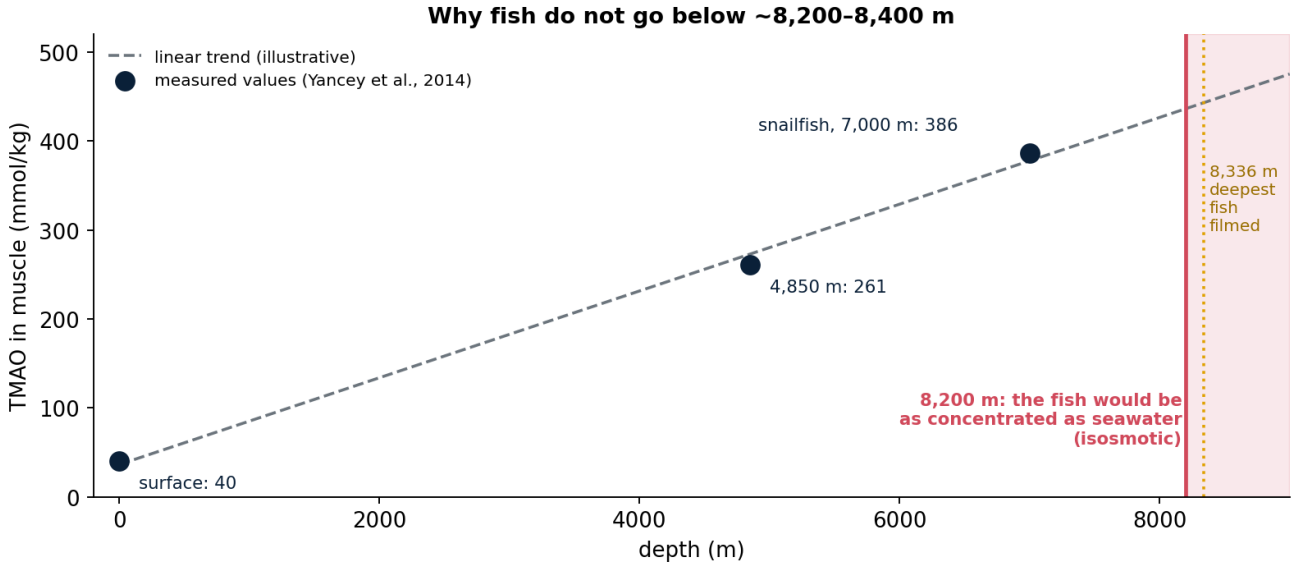
Key study: Yancey et al., *PNAS* 111, 4461–4465 (2014). **[V]** (0.9)

1. TMAO increases with depth: from 40 to 261 mmol/kg between the surface and 4,850 m.
2. In snailfish caught at 7,000 m in the Kermadec Trench they measured 386 ± 18 mmol/kg, the highest value recorded.

3. More TMAO means more solutes in the body. A shallow-water fish has about 350 mOsmol/kg; seawater, about 1,100. The 7,000 m snailfish were already at 991.
4. Extrapolating, at **8,200 m** the fish would be **isosmotic**: as concentrated inside as the sea outside.
5. Deeper still it would be more concentrated than the sea and water would flood into its body. It would have to reverse its entire salt-regulation system, which works the other way round.

This is the first evidence that biochemistry, rather than lack of food or oxygen, limits the depth of an entire animal group.

- **Isosmotic** (*ísos* "equal"): having the same solute concentration as the surroundings.
- **Osmoregulation**: control of the body's water and salt balance.



Original figure from this study · CC BY 4.0

Figure 15. TMAO increases with depth. At around 8,200 m the fish would match the concentration of seawater; the deepest fish ever filmed is right at that limit.

11.4 Invertebrates play by other rules

Marine invertebrates are already isosmotic with seawater, so they do not have that ceiling. Amphipods also increase TMAO with depth, but they replace other osmolytes (such as taurine) with glycerophosphocholine and similar compounds. Both fish and amphipods also accumulate **scyllo-inositol**, a compound that breaks up aggregates of misfolded proteins. **[V]** (0.75)

11.5 Other adaptations [C] (0.85)

- Membranes with more unsaturated fats, which stay fluid under pressure.
- Proteins with modified sequences that resist compression.
- No gas-filled swim bladder.
- Poorly calcified bones and a gelatinous body.
- Slow metabolism and long life. In hadal amphipods, the low turnover rate of tissues is documented. **[V]**

12. Trenches as sinks for carbon and pollution

12.1 Hotspots of microbial activity

Study: Glud et al., *Nature Geoscience* 6, 284–288 (2013). **[V]** (0.9)

- They measured sediment oxygen consumption in situ at Challenger Deep with an autonomous micro-profiler.
- Result: biological oxygen consumption was **twice** that at a nearby 6,000 m site.
- There were more microbial cells in the sediment at the bottom of the trench.
- Lead-210 profiles (^{210}Pb, a natural radioactive isotope used as a clock to measure how much sediment is deposited) indicated high sedimentation.
- Conclusion: the elevated supply of organic matter sustains intense microbial activity despite the pressure.

Key idea: **deeper does not mean less life**. A trench works like a hole in a garden: everything that blows over it ends up falling in.

Later data from the Danish Center for Hadal Research confirm the pattern in several trenches **[V]** (0.8):

- Hadal sediments show biological activity **2 to 8 times higher** than neighboring abyssal sediments (Mariana, Tonga, Kermadec, Atacama and Puysegur).
- The mean carbon burial rate along trench axes is **about 70 times** the deep-sea average, per square meter.
- Earthquake-triggered sediment slides have increased the accumulation of organic carbon over the last 150 years by up to a factor of 4.
- Hadal sediment is enriched in **mercury** 20 to 50 times relative to the deep-sea average.

12.2 Persistent organic pollutants

Study: Jamieson et al., *Nature Ecology & Evolution* (2017). **[V]** (0.9)

- They analyzed amphipods from the Mariana and Kermadec trenches, 7,000 km apart, caught as deep as 10,250 m.
- They found **PCBs** and **PBDEs** in all samples, all species and all depths in both trenches.
- The levels were comparable to those of Suruga Bay (Japan), one of the most polluted industrial zones of the northwest Pacific.
- In the Mariana Trench, the highest PCB values were 50 times greater than in crabs from paddy fields fed by the Liaohe River, one of the most polluted in China.
- Proposed route: contaminated plastic debris and sinking carcasses.

Terms:

- **POPs** (persistent organic pollutants): synthetic compounds that do not break down for decades.
- **PCBs** (polychlorinated biphenyls): electrical insulators; about 1.3 million metric tons were made between the 1930s and their ban in the 1970s.
- **PBDEs** (polybrominated diphenyl ethers): flame retardants.
- **Bioaccumulation**: a pollutant builds up in tissues (especially fat) faster than it is eliminated, and increases up the food chain.

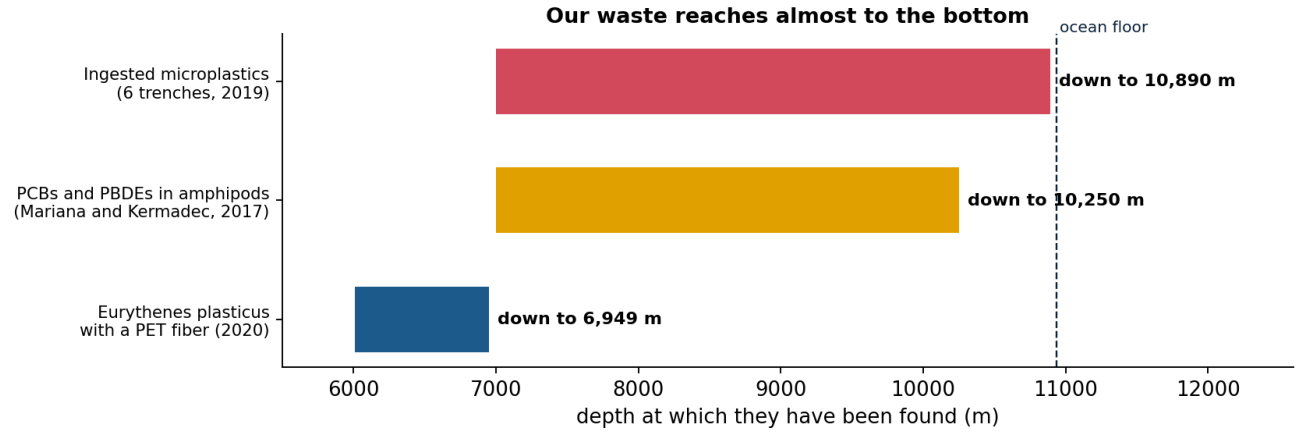
12.3 Carbon from atomic bombs

Study: Wang et al., *Geophysical Research Letters* (2019). **[V]** (0.85)

The nuclear tests of the 1950s and 1960s temporarily doubled the carbon-14 (^{14}C, a radioactive isotope of carbon) in the atmosphere. That "bomb carbon" has been detected in amphipods from the Mariana, Mussau and New Britain trenches. It shows that surface carbon reaches the bottom within a few decades, far faster than ocean circulation, through the food chain.

12.4 Plastics

- **2019** (Jamieson et al.): microplastics ingested by amphipods in six Pacific trenches (Japan, Izu–Bonin, Mariana, Kermadec, New Hebrides and Peru–Chile), between 7,000 and 10,890 m. It is the deepest record of microplastic ingestion. **[V]** (0.9)
- **2020**: ***Eurythenes plasticus*** is described, an amphipod new to science collected between 6,010 and 6,949 m in the Mariana Trench. One specimen had in its gut a fiber 83.74% similar to **PET** (polyethylene terephthalate, the plastic of drinks bottles). It was the first new species described already contaminated with plastic. **[V]** (0.9)



Original figure · the shallow end of the PCB bar is approximate; the maximum (10,250 m) is the published value · CC BY 4.0

Figure 16. Depths at which pollutants of human origin have been found in hadal fauna.

12.5 Conclusion of the section

Trenches are not isolated. The same funnel that concentrates food concentrates our waste, and the organisms that live there are long-lived and slow-metabolizing, which favors accumulation.

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13. What has NOT been demonstrated

13.1 "Life entirely independent of the Sun"

Mostly false. (0.85)

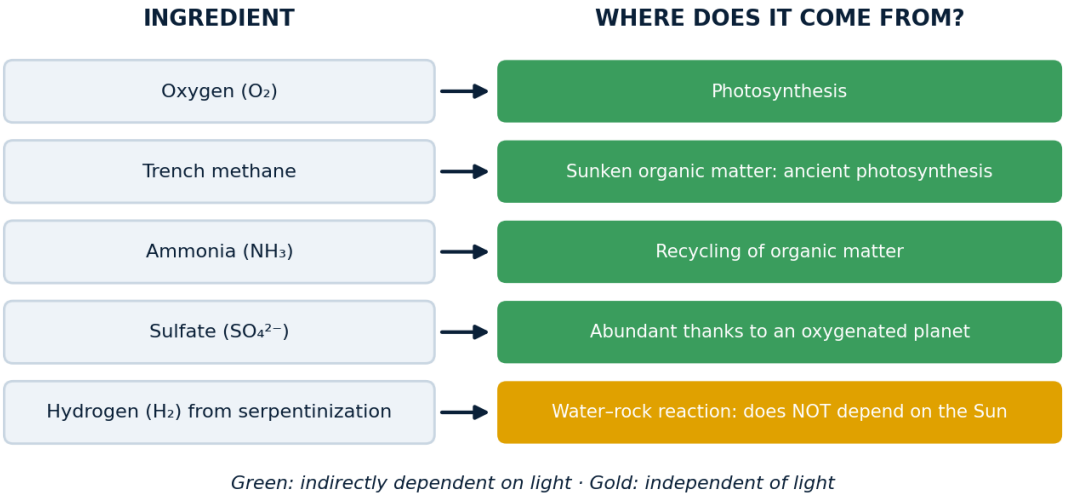
Ingredient	Where does it really come from?
Oxygen (reactions 1, 2, 6, 7)	Photosynthesis
Trench methane	Organic matter sunk from the surface: ancient photosynthesis
Sulfate (reaction 3)	Its abundance in the sea depends on an oxygenated planet

Ingredient	Where does it really come from?
Ammonia (reaction 6)	Recycling of organic matter

Real exceptions: methanogenesis and acetogenesis (reactions 4 and 5) when the **H_2 is of geological origin**, as in serpentinization. There neither the energy nor the carbon depends on light. That is why Lost City is of such interest to those who study the origin of life.

Even at Shinkai Seep Field, at 5,700 m and on mantle rock, part of the fauna uses material of photosynthetic origin. **[V]**

Correct wording: these ecosystems do not need light on site, but almost all of them depend on products of photosynthesis generated somewhere else or at some other time.



Original figure from this study · CC BY 4.0

Figure 17. Where each ingredient of chemosynthesis comes from. Only geological hydrogen is independent of light.

13.2 "Dark oxygen" produced by metallic nodules

Strongly disputed. Confidence that the phenomenon is real: 0.2.

- **What was claimed.** Sweetman et al. (*Nature Geoscience* 17, 737–739, 2024): in sealed chambers on the abyssal Pacific seafloor covered with **polymetallic nodules**, oxygen rose over two days to more than three times the background value. They measured voltages of up to 0.95 V on nodule surfaces and proposed that these might split water by **electrolysis**. **[V]**
- **Criticisms.** An initial oxygen rise is a known artifact of these experiments (trapped air bubbles); chambers without nodules also showed increases; one figure attributed production to an area where there were no nodules in the chamber, a point the lead author conceded; the measured voltages are not sufficient for electrolysis; there is no elevated oxygen in the water above the region; another research group, using the same landers, attributed a similar result to bubbles. **[V]**
- **Conflict of interest.** The loudest rebuttal comes from a mining company which had also funded part of the original study. But there are independent criticisms too.
- **Current status.** Since **8 April 2026** the article has carried an editor's note alerting readers that aspects of the paper are subject to concerns being considered by the editors. **[V]** The original team announced a three-year project to try to reproduce the result. Developments in 2026 are covered in section 17.2, question 9.

In addition, it concerns abyssal plains (~4,000 m), not trenches. It should not be presented as fact.

Terms:

- **Polymetallic nodule**: a potato-sized concretion rich in manganese, nickel and cobalt that grows millimeters per million years on the abyssal seafloor.
- **Electrolysis** (*lýsis* "loosening"): decomposition of a substance by electric current; here, of water into H_2 and O_2.
- **Benthic chamber**: a container pushed into the seafloor to isolate a patch of sediment and measure how the chemistry of the enclosed water changes.
- **Experimental artifact**: a result produced by the measuring method and not by the phenomenon being studied.

13.3 "A global chemosynthetic corridor"

A reasonable hypothesis, based on two neighboring trenches in the northwest Pacific. Equivalent exploration of other trenches is lacking. (0.5)

13.4 Figures in circulation without a solid basis

- Area of the hadal zone: between under 0.2% and 2% depending on the source (see 3.2).
- "Exact" depth of Challenger Deep: varies with method and year.
- Total abundances of fauna in a trench: unsupported extrapolations.

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14. Implications: climate, mining and astrobiology

14.1 Climate

- Anaerobic oxidation of methane removes around 90% of the methane in marine sediments. It is a biological filter against a powerful greenhouse gas. **[V]** (0.85)
- If there are large reserves of methane and hydrates beneath trenches, it matters how stable that store is and how efficient the filter is.
- Trenches bury organic carbon efficiently and, through subduction, can carry it into the Earth's interior: a sink on a geological timescale. **[C]** (0.7)

14.2 Deep-sea mining

- Polymetallic nodules lie on abyssal plains, not in trenches, but the debate is the same: there is a wish to exploit a seafloor that is barely known.
- In 2025 it was found that almost 90% of the hadal microbes sampled were unknown. The impact on what has not been cataloged cannot be assessed.

14.3 Biotechnology

- The cellulase of *Hirondellea gigas* produces glucose from wood, straw or paper; it is of interest for biofuels. **[V]**
- The MEER authors highlight the potential of hadal microbes as a source of new genes, structures and functions. **[V]**

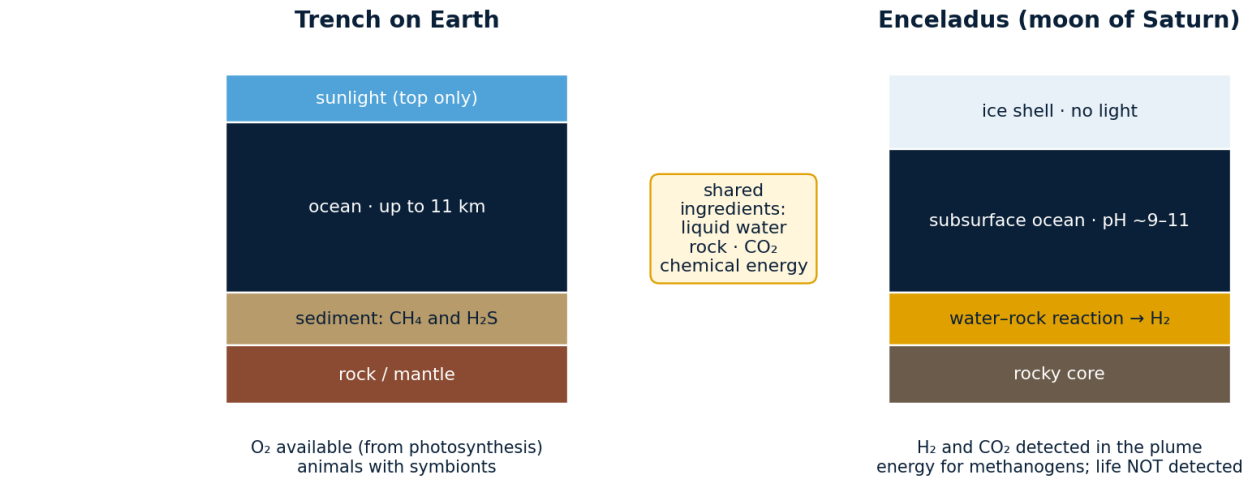
14.4 Astrobiology

Astrobiology: the study of the origin, evolution and possible distribution of life in the universe.

Enceladus (moon of Saturn) **[V]** (0.9):

- It has an ocean beneath an ice shell and a **plume** of material escaping from the south pole.
- On 28 October 2015 the *Cassini* spacecraft flew through the plume 49 km above the surface. Its **mass spectrometer** (INMS, an instrument that identifies molecules by their mass) detected **molecular hydrogen**.
- Publication: Waite et al., *Science* 356, 155–159 (2017).
- The plume contained up to 1.4% H_2 and up to 0.8% CO_2 by volume: the two ingredients of methanogenesis (reaction 4).
- The most plausible interpretation is that the hydrogen comes from hydrothermal reactions between water and rock, as in serpentinization.
- The ocean would be alkaline (pH ~9–11), like the fluids of Lost City.
- **Life was not detected.** What was detected is that chemical energy is available for a metabolism like that of methanogens.

Trenches and Lost City are the best terrestrial analogues. And the lesson of section 13.1 applies: on a world without photosynthesis there would be no oxygen, so the life to be expected would resemble methanogens, not tubeworms.



Original schematic, not to scale · CC BY 4.0

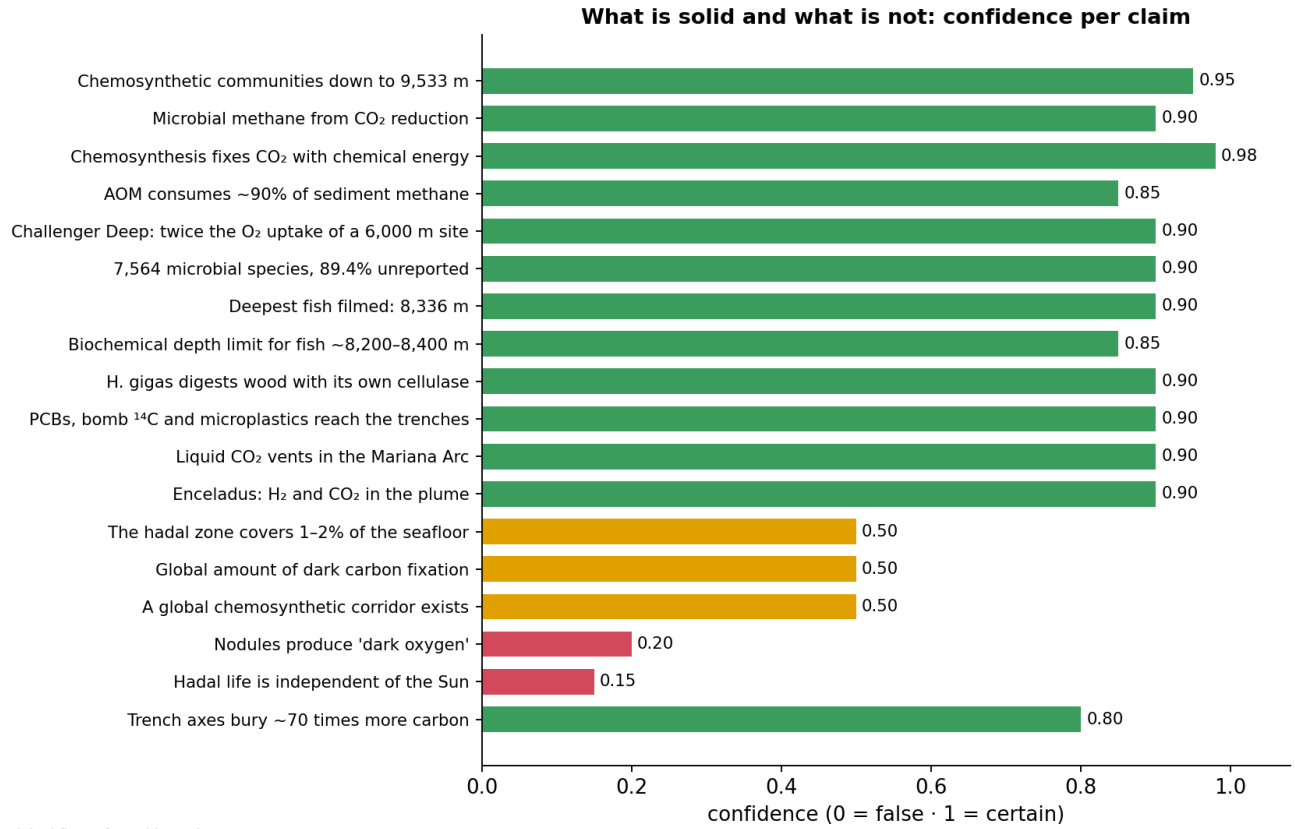
Figure 18. Schematic comparison between a trench on Earth and the subsurface ocean of Enceladus.

15. Verification table and biases

Method: (1) split the topic into claims; (2) assign confidence; (3) check facts and look for biases; (4) combine; (5) state the limits.

#	Claim	Confidence	Status
1	There are chemosynthetic animal communities down to 9,533 m	0.95	Verified (*Nature* 2025)
2	Their methane is microbial, from CO_2 reduction	0.9	Verified (*Nature* 2025)

#	Claim	Confidence	Status
3	Chemosynthesis fixes CO_2 with chemical energy	0.98	Consensus
4	AOM consumes ~90% of the methane in marine sediments	0.85	Verified (2025 review)
5	Challenger Deep consumes twice as much oxygen as a 6,000 m site	0.9	Verified (*Nat. Geosci.* 2013)
6	7,564 hadal microbial species, 89.4% new	0.9	Verified (*Cell* 2025)
7	Deepest fish filmed: 8,336 m	0.9	Verified
8	Biochemical limit for fish ~8,200–8,400 m	0.85	Verified (*PNAS* 2014); it is an extrapolation
9	*H. gigas* digests wood with its own cellulase	0.9	Verified (*PLoS ONE* 2012)
10	PCBs, bomb ^{14}C and microplastics reach the trenches	0.9	Verified
11	Liquid CO_2 vents in the Mariana Arc	0.9	Verified (2006)
12	Enceladus has H_2 and CO_2 in its plume	0.9	Verified (*Science* 2017)
13	The hadal zone covers 1–2% of the seafloor	0.5	Contradictory sources
14	Global dark carbon fixation	0.5	Estimates differ by up to a factor of 10
15	A global chemosynthetic corridor exists	0.5	Hypothesis
16	Nodules produce "dark oxygen"	0.2	Disputed; editor's note since April 2026
17	Hadal life is independent of the Sun	0.15	Incorrect simplification
18	Trench axes bury ~70 times more carbon than the deep-sea average	0.8	Verified (presentation by R. Glud, 2026)



Original figure from this study · CC BY 4.0

Figure 19. The 18 claims in the table, in the same order. Green: solid. Gold: doubtful. Red: not supported.

Biases detected:

- **Novelty.** The 2025 headlines spoke of "life never seen before". Hadal chemosynthesis has been known since 1999; what is new is the depth, the abundance and the extent.
- **Interested source.** In the dark-oxygen debate both the authors and the critical mining company have incentives.
- **Sampling.** Only a tiny fraction of the hadal seafloor has been visited directly. Any generalization is provisional.
- **Geographical and institutional.** The major 2025 results come from a single program and a single submersible. They appeared in top journals, but independent replication is lacking.
- **Press release.** Some figures (number of dives with communities, size of the fields) come from press releases or media reports, not from the paper; they have been given lower confidence.

16. Glossary of technical terms

In alphabetical order. The origin of the word is given where it helps to remember it (Gr. = Greek, Lat. = Latin).

Term	Origin	Meaning
Abiotic	Gr. *a-* "without" + *bíos* "life"	Produced without the involvement of living things
Abyssal	Gr. *ábyssos* "bottomless"	Zone between 4,000 and 6,000 m; the great plains of the ocean floor

Term	Origin	Meaning
Accretion (accretionary prism)	Lat. *accretio* "growth"	Wedge of sediment piled up at the edge of a subduction zone
Accretionary wedge		Another name for the accretionary prism
Acetogenesis		Microbial production of acetate from CO_2 and H_2
Aerobic / anaerobic	Gr. *aér* "air" + *bíos*	Using oxygen / living without it
Amphipod	Gr. *amphí* "both" + *poús* "foot"	Small shrimp-like crustacean; dominant scavenger in trenches
ANME	acronym	Archaea that oxidize methane without oxygen
AOA / NOB	acronyms	Ammonia-oxidizing archaea / nitrite-oxidizing bacteria
AOM	acronym	Anaerobic oxidation of methane
Archaea	Gr. *archaîos* "ancient"	Microorganisms without a nucleus, in a domain separate from bacteria
Artifact		A false result caused by the measuring method
Astrobiology	Gr. *ástron* "star"	Study of life in the universe
Authigenic	Gr. *authigenés* "born in place"	Mineral formed in the place where it is found
Autotroph	Gr. *autós* "self" + *trophé* "food"	Organism that makes its organic matter from CO_2
Bacteriocyte		Animal cell specialized in housing symbiotic bacteria
Bathyal	Gr. *bathýs* "deep"	Zone between 1,000 and 4,000 m
Bathymetry	Gr. *bathýs* + *métron* "measure"	Measurement of ocean depths
Benthic	Gr. *bénthos* "bottom"	Living on or in the seafloor
Benthic chamber		Container that isolates a patch of seafloor to measure its chemistry
Bioaccumulation		Progressive build-up of a pollutant in tissues
Biogenic		Produced by living things
Bioluminescence	Lat. *lumen* "light"	Light produced by a living organism
Bivalve	Lat. *bi-* "two" + *valva* "door leaf"	Mollusk with a two-part shell
Black smoker		Hydrothermal chimney emitting dark, acidic, very hot fluid
Carbonate		Mineral with the CO_3^{2-} group; the commonest is calcium carbonate (limestone)
CCD	acronym	Carbonate compensation depth; below it, calcium carbonate dissolves
Cellulase		Enzyme that breaks down cellulose
Chemoautotroph		Organism that fixes CO_2 with chemical energy
Chemosynthesis	Gr. *chemeía* + *sýnthesis*	Production of organic matter with chemical energy
Clathrate	Lat. *clathratus* "latticed"	See *Gas hydrate*

Term	Origin	Meaning
Cold seep		Slow escape of methane- and sulfide-bearing fluids at ambient temperature
Consortium		Association of microbes cooperating in a reaction
$\delta^{13}C$		Measure of the ratio of heavy to light carbon, in ‰
δD		Measure of the ratio of deuterium (heavy hydrogen) to light hydrogen, in ‰
Décollement	French "detachment"	Sliding surface between the sinking plate and the plate above it
Deep-sea gigantism		Tendency of certain deep invertebrates to be very large
Diagenesis	Gr. *diá* "through" + *génesis*	Chemical and physical changes in sediment after burial
DIC	acronym	Dissolved inorganic carbon: CO_2 + bicarbonate + carbonate
Echo sounder		Device that measures depth from the echo of a sound pulse
Electrolysis	Gr. *lýsis* "loosening"	Decomposition of a substance by electric current
Electron acceptor		Substance that receives electrons in a redox reaction; the "oxidant" (O_2, sulfate, CO_2)
Electron donor		Substance that gives up electrons; the "fuel" (H_2S, CH_4, H_2)
Endemic	Gr. *éndemos* "native"	Living in one particular place only
Endosymbiont	Gr. *éndon* "within"	Organism living inside the cells or tissues of another
Enzyme	Gr. *en* "in" + *zýme* "leaven"	Protein that speeds up a chemical reaction
Epipelagic	Gr. *epi* "upon"	Sunlit surface layer (0–200 m)
Extremophile	Lat. *extremus* + Gr. *phílos* "friend"	Organism that thrives in extreme conditions
Fault		Fracture in the crust with displacement of blocks
Fixation (carbon)		Conversion of CO_2 into organic compounds
Forearc		Region between the trench and the volcanic arc
Frenulate	Lat. *frenulum* "little bridle"	Subgroup of very thin siboglinids
Gas hydrate		Ice-like solid with gas molecules trapped in cages of water
Gb / Tbp		Gigabases / terabase pairs: a billion / a trillion "letters" of DNA
Hadal	Gr. *Hades*	Zone below 6,000 m; almost all of it in trenches
Heterotroph	Gr. *héteros* "other"	Organism that needs ready-made organic matter
Holotype / paratype	Gr. *hólos* "whole", *pará* "beside"	The single specimen that serves as the official reference for a species / the other specimens used in describing it
HOV / ROV	acronyms	Human-occupied vehicle / remotely operated vehicle on a cable

Term	Origin	Meaning
Hydrogenotrophic		Using H_2 as an energy source
Hydrothermal	Gr. *hýdor* "water" + *thermós* "hot"	Relating to water heated by the Earth's interior
Ikaite	from Ikka Fjord, Greenland	Hydrated calcium carbonate, stable only in very cold water
Isosmotic	Gr. *ísos* "equal"	Having the same solute concentration as the surroundings
Isotope	Gr. *ísos* + *tópos* "place"	Variant of an element with a different number of neutrons
Lander		Autonomous platform dropped to the seafloor that returns on its own
Lithotroph	Gr. *líthos* "stone"	Obtaining electrons from inorganic compounds
Mantle		Layer of the Earth beneath the crust
Marine snow		Rain of organic particles toward the seafloor
Metagenomics	Gr. *metá* "beyond"	Sequencing all the DNA in an environmental sample
Methanogenesis / methanogen		Biological production of methane / microbe that does it
Methanotrophy / methanotroph		Use of methane as food / organism that does it
Microplastic		Plastic fragment smaller than 5 mm
Mid-ocean ridge		Submarine mountain range where new crust is created
MPa		Megapascal; 1 MPa ≈ 10 atmospheres
Mutualism		Symbiosis in which both partners benefit
Nitrification		Biological oxidation of ammonia to nitrite and of nitrite to nitrate
Normal fault		Fault in which one block drops relative to the other as the crust stretches
Oligotrophic	Gr. *olígos* "little" + *trophé*	Nutrient-poor
Olivine		Green iron–magnesium mineral, main component of the upper mantle
Osmolyte	Gr. *osmós* "push"	Molecule a cell accumulates to regulate its water
Osmoregulation		Control of water and salt balance
PBDEs / PCBs	acronyms	Flame retardants / electrical insulators; persistent pollutants
Pelagic	Gr. *pélagos* "open sea"	Relating to the water column
Peridotite		Mantle rock, rich in olivine
PET	acronym	Polyethylene terephthalate; plastic of bottles and textile fibers
Pg / Tg		Petagram (10^{15} g) / teragram (10^{12} g)
Photopic (vision)	Gr. *phôs* "light"	Vision in bright light
Piezolyte		Osmolyte that protects proteins from pressure

Term	Origin	Meaning
Piezophile	Gr. *piézein* "to press" + *phílos*	Organism adapted to high pressure
Plume		Jet of material; also the feathery gill crown of a tubeworm
Polychaete	Gr. *polýs* "many" + *chaíte* "bristle"	Bristled marine annelid worm
Polymetallic nodule		Concretion rich in manganese, nickel and cobalt on the abyssal seafloor
POPs	acronym	Persistent organic pollutants
Prokaryote	Gr. *pró* "before" + *káryon* "kernel"	Cell without a nucleus
Redox		Reaction involving a transfer of electrons
Reduced / oxidized		Electron-rich / electron-poor
Scavenger		Animal that eats carcasses
Serpentinization	Lat. *serpens*	Water–peridotite reaction that produces H_2, heat and alkaline fluids
Siboglinid		Family of mouthless tubeworms that live on symbionts
16S rRNA		Gene used as a "barcode" to identify bacteria and archaea
SRB	acronym	Sulfate-reducing bacteria
Subduction	Lat. *sub* + *ducere*	Sinking of one tectonic plate beneath another
Symbiosis	Gr. *sýn* "with" + *bíos*	Close association between two species
Syntrophy	Gr. *sýn* + *trophé*	Obligate metabolic cooperation between two microbes
Teleost	Gr. *téleios* "complete" + *ostéon* "bone"	Modern bony fish; almost all living fish are teleosts
Thaumarchaeota	Gr. *thaûma* "wonder"	Group of ammonia-oxidizing archaea, very abundant in the ocean
Thermogenic		Produced by heat at great depth
TMAO	acronym	Trimethylamine N-oxide; main piezolyte in fish
Trophosome	Gr. *trophé* + *sôma* "body"	Organ of siboglinids that houses the bacteria
Xenophyophore	Gr. *xénos* "foreign"	Giant single-celled organism of the deep seafloor

17. Limitations, open questions and conclusions

17.1 Limitations of this report

- It is a **synthesis of the published literature**. It is not original research and has not been peer-reviewed.
- Prepared with AI assistance (Claude, by Anthropic).

- Data marked **[V]** were checked against published sources in October 2026. Only the paper on *Pseudoliparis swirei* was read in full; for the rest, abstracts, figure captions, press releases or excerpts were consulted.
- Data marked **[C]** come from general knowledge and should be checked before being cited in academic work.
- The etymologies in the glossary are indicative and simplified.
- The chemical equations are simplified formulations; real metabolism has many intermediate steps.
- This English version is a translation of the Spanish original; in case of discrepancy the Spanish text prevails.

17.2 The nine open questions and what science says today

For each question, the current state of knowledge is given according to the sources consulted in October 2026.

1. Are there communities like those of 2025 in other trenches?

Status: unconfirmed. No equivalent find has been published for the Mariana, Tonga, Kermadec or Peru–Chile trenches. What is known:

- Japan Trench: clam communities between 7,326 and 7,434 m.
- Mariana forearc: Shinkai Seep Field, at 5,700 m, on serpentinite.
- Mariana Trench, Sirena Deep: possible microbial mats at about 10,700 m, with no animals.
- Geological maps show similar faults in the Kermadec, Tonga and Peru–Chile trenches, which are cited as targets for future cruises.

Confidence that they will eventually be found in more trenches: 0.6 (a reasonable inference, not a datum).

2. How much methane and hydrate lies beneath trenches, and how stable is it?

Status: not quantified. No published estimate of the total amount was found. What is established is qualitative: the methane is microbial, the pressure and temperature conditions fall within the hydrate stability field (Plate 3), and the geological model proposes hydrate lenses between the layers (Plate 4). The authors speak of the deepest known "methane reservoir". There are no data on its long-term stability. (0.8)

3. What fraction of the diet of "ordinary" hadal fauna comes from chemosynthesis?

Status: not quantified; there are indications. **[V]** (0.7)

- Isotope analyses of animals from the Mariana and Kermadec trenches indicate **several food sources** at once and highly dynamic food webs, but do not separate out the share due to chemosynthesis.
- A study of bacterial lipids in the Mariana and Yap trenches suggests that chemoautotrophic carbon fixation in the subduction zone could be an effective source of organic carbon on the seafloor.
- At Shinkai Seep Field the community relies mostly on chemosynthesis, but that is an oasis, not the "ordinary" seafloor.

4. How do larvae colonize oases hundreds of kilometers apart?

Status: partly answered for hydrothermal vents; unknown for hadal seeps. **[V]** (0.75)

- Adults are fixed in place; only the larvae travel, carried by deep currents.

- After an eruption wiped out a community in the eastern Pacific, the site was recolonized by larvae arriving from distant sites, of species different from the previous ones.
- On the Mid-Atlantic Ridge, mussel larvae do not usually drift more than 150 km: connection between distant fields happens over generations, via intermediate habitats that nobody has yet located ("phantom stepping stones").
- The accepted model is that of **stepping stones**: each oasis receives larvae from its neighbors.
- There are no studies for trenches. That 19 of 23 dives found communities along 2,500 km suggests that the stepping stones there are very close together; this is an inference of this report, not a published result.

5. Is there chemosynthetic life below 9,533 m?

Status: animals, none known; microbes, probably yes. **[V]** (0.75)

- The DEEPSEA CHALLENGE expedition (2012) photographed, at Sirena Deep at about 10,700 m, rocks covered with filaments interpreted as microbial mats. The study, published in 2020, proposes that they live on compounds produced when the rock reacts with water.
- In the bottom waters of Challenger Deep and Sirena Deep, microbes able to use sulfur, methane and hydrogen have been detected.

6. What do the thousands of new microbial species do?

Status: answered only in broad outline. **[V]** (0.75)

- Two general strategies are known (Figure 20), along with prominent functions: degradation of aromatic compounds and antioxidant defense.
- In waters below 10,000 m, genes for an unusual nitrogen metabolism are abundant: conversion of nitrate to ammonium, nitrogen fixation and urea transport.
- For the great majority only the genome exists. They have not been cultured, so their real function remains untested.

7. How much carbon does the dark ocean really fix?

Status: unresolved. Estimates range from 0.1 to 1.2 Pg C per year depending on the method (section 8.7 and Figure 12). The discrepancy is attributed to energy sources that the budgets do not account for. (0.7)

8. What effect do pollutants have on such long-lived animals?

Status: unknown. Accumulation is documented (PCBs and PBDEs in amphipods, mercury in sediment, microplastics in guts), but no study measuring harm to hadal fauna was found. The authors of the 2017 work themselves noted that they did not know what it meant for the ecosystem. (0.8)

9. Is oxygen produced without light on the seafloor?

Status: not confirmed. **[V]** (0.85)

- No other study has observed the phenomenon independently; earlier measurements in nodule areas showed only oxygen consumption.
- A reanalysis published in 2026 concludes that the correlation between nodule surface area and oxygen produced, which was the main evidence, disappears when all the chambers are included: there would be no demonstrated link between nodules and oxygen.
- A German team working in the same region stated that its data show no sign of oxygen production.

- The original team stands by its result. In January 2026 it presented two new landers designed to measure seafloor respiration and announced a cruise to the Clarion–Clipperton Zone for spring 2026, funded by the Nippon Foundation.
- No published results from that cruise were found as of the date of this report.

17.3 Conclusions

1. **The hadal zone is an active system.** It receives, transforms and buries carbon far more intensely than the plains around it.
2. **Chemosynthesis is a significant energy source in trenches**, not a curiosity. This is shown by the 2025 communities, the isotopic signature of the methane and the geology that feeds them.
3. **CO_2 is the chemical link** between the organic matter that sinks, the methane that forms below the seafloor and the biomass that grows at the seeps.
4. **Independence from the Sun is the exception.** Only systems driven by hydrogen of geological origin achieve it, which is why they are the most interesting analogs in the search for life beyond Earth.
5. **Knowledge is recent and partial.** The most striking data come from a single program and two trenches; none of the nine open questions has a complete answer yet.
6. **The human footprint is already there.** Trenches are accumulating pollutants before they have been explored.

Recommendations for anyone using this report:

- Always cite the primary source, not this document, for any numerical value.
- Treat claims with confidence below 0.8 as provisional.
- Check the status of questions 1, 2 and 9, which are the most likely to change in the short term.

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18.1 Figures 1 to 23: original

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Figures	Type	Source data
1, 2, 4–8, 10, 17, 18	Explanatory schematics	Own work
3	Charts	WHOI-HADES, MEER portal and the measurements cited in 3.4
9	Chart	Lupton et al. 2006; Kelley et al. 2005
11	Chart	Measured values from the cited sources; typical ranges from general knowledge
12	Chart	Estimates cited in 8.7
13	Chart	Fujikura et al. 1999; Onishi et al. 2018; Peng et al. 2025
14, 20	Charts	MEER project, *Cell* 2025

Figures	Type	Source data
15, 21	Chart and drawing	Yancey et al. 2014; records cited in 10.3
16	Chart	Jamieson et al. 2017 and 2019; Weston et al. 2020
19	Chart	Table in section 15
22	Chart	Data from sections 9 and 10
23	Chart	Jamieson (2015), *The Hadal Zone*

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Citation: Peng, X., Du, M., Gebruk, A. et al. *Flourishing chemosynthetic life at the greatest depths of hadal trenches*. Nature 645, 679–685 (2025). https://doi.org/10.1038/s41586-025-09317-z. License: https://creativecommons.org/licenses/by-nc-nd/4.0/

18.4 Plates 5 and 6: *Zootaxa*, CC BY 3.0

From Gerringer, M. E., Linley, T. D., Jamieson, A. J., Goetze, E. and Drazen, J. C. (2017). *Pseudoliparis swirei* sp. nov.: A newly-discovered hadal snailfish (Scorpaeniformes: Liparidae) from the Mariana Trench. *Zootaxa* 4358 (1), 161–177. https://doi.org/10.11646/zootaxa.4358.1.7

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18.5 Images reviewed and not included

Image	Reason
Poseidon chimney, Lost City (2005), https://noaa.gov/media/66837	Credited "courtesy of University of Washington". Permission has been requested; it will not be used without it
Video of the snailfish at 8,336 m (University of Western Australia)	All rights reserved
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19. References

Only links consulted during preparation are included.

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